胃腸肝膽科系病房工作手冊

2021 年 11 月製作

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胃腸肝膽科系工作規範與須知

- 請準時參加晨會：07:30 在 MICU6 討論室舉行。請注意教學門診時間，務必出席。

臨床照護

1. 病患 service 由總醫師調整以符合臨床工作量(PGY2 ≤14 床，PG1 ≤10 床)。
2. 平日臨床照護工作時間至 16:30。16:00 後入院之新病人由值班人員負責。
3. 接新病人時請看門診/急診病歷與住院通知單。於查房後知會門診主治醫師並詢問診療計畫。
4. 每日依臨床狀況評估藥物，週五請 renew 藥物。病患有感染徵象：先留檢體，再給抗生素。
5. 遇患者病情變化 (意識改變、生命徵象不穩、出血)，請找總醫師幫忙。
6. 侵入性處置請找資深醫師指導並維持無菌操作，不應遺留的東西 (針頭、guide wire) 務必取出。
7. 有臨床需求可到 MICU6 借超音波：填寫登記本、小心使用、儘速歸還。

病歷紀錄

1. Progress note : 每日紀錄病情變化、檢查、治療、抗生素及 VS comment。
2. Weekly summary : 每週五，及連續假期前書寫，以便值班醫師了解病情及計畫。
3. Duty note : 遇突發狀況 (生命徵象變化/自動離院/死亡) 時請紀錄。CPCR 請登錄急救紀錄。
4. Transfer note : 病患轉加護病房或轉科時，請寫 transfer note 並與對方住院醫師/總醫師交班。
5. 每日下班前請電子交班，不穩定病患請另口頭交班(與值班醫師及總醫師)。

出院準備

- 預期出院請於前一天開醫囑。臨時出院請於 10:00 前開醫囑。與主治醫師確認回診時間/抽血。

肝癌治療，肝炎治療

- 化療前檢驗 : CBC/DC, liver function。向主治醫師確認病人接受的 regimen 及劑量。
- 免疫治療請 : (1) 知會門診主治醫師 (2) 確認藥物是否自費 (3) 完成給藥前的檢驗。
- B 肝治療請確認用藥及劑量。聯絡肝病中心 (分機 2269) 告知使用條件及主治醫師後開立。

  C 肝治療藥物選擇請直接與主治醫師討論後，告知總醫師。

HBV DNA 定量檢驗	M23-078	HCV RNA 定量檢驗	M23-079

常用診斷

肝原發惡性腫瘤/肝細胞癌	C228/C220	膽囊惡性腫瘤	C23
肝內膽管癌/未明膽管癌	C221/C249	胰臟頭部/體部惡性腫瘤	C250/C251
肝、膽囊及膽道性態未明腫瘤	D376	十二指腸壺腹惡性腫瘤	C241
胃惡性腫瘤	C169	肝硬化	K7460

安排檢查可由系統右上角 工具箱 >檢查/檢驗查詢 以科別搜尋，或直接輸入代碼搜尋。同名的檢查可出現在不同專科，輸入代碼時請確認完全一致(連同字母)

- 完成醫囑請電聯科內檢查室確認時間，說明：(1) 病人資料 (2) 檢查項目 (3) 是否減痛。
- 侵入性檢查前：完成同意書並請家屬陪同。注意禁食時間。請停用抗凝血/抗血小板藥物。
  檢查前檢驗 Hb、PT、aPTT、platelet。備血，並矯正凝血功能 (INR <1.5, platelet > 60k)。
- TACE, RT, HAIC 前詢問主治醫師是否停用 TKI，停用多久。

影像診療科

(櫃檯 2570、電腦斷層 2585 # 316、血管攝影 2585 # 323/324、急診電腦斷層 2568)

Fluoroscopy	X75-981	食道氣球擴張、放 N-D tube 請開 M23-002 + X75-981
Liver CT	X75-992	如需指定醫師判讀請選病理大樓檢查室。IVF: prevent AKI
Liver MRI	X75-989	如需指定曾振輝醫師判讀請排週一或週六檢查
CT-guided biopsy	X75-983	CT-guided biopsy of abdomen
TACE of liver	X75-896	藥物：化療醫囑 → C220 → TACE: doxyrubicin 20mg

核子醫學科 (B1 檢查室 2620：bleeding scan, bone scan、1F 檢查室 3596：whole body scan)

Bleeding scan (N73-037)、Bone scan (N73-060)、Whole body tumor scan(N73-061A)

GI 常用檢查.

*** 確認是否需鎮靜　# 確認是否需指定醫師**

內視鏡室（分機 2227, 2228）
治療中心（分機 2217, 2221）

胃鏡　*	M23-002	Panendoscopy (Esophagogastroduodenoscopy)
POEM　* # EMR/ESD　* #	M23-002	+ 全身麻醉：會診麻醉科，請 CR 輸入排程 + 備 LPRBC & FFP & platelet. Keep ivf, 確認自費 EMR/ESD 同意書：消化道內視鏡粘膜下層剝離-粘膜切除。 食道 RFA 同意書：消化道內視鏡檢查
食道 RFA　* #	M23-085	
EV ligation	M23-009	Esophageal varices ligation
GV sclerotherapy	M23-081	Gastric variceal sclerosing therapy
膠囊 (NT$ 60-70k)	M23-082	Capsule endoscopy
小腸鏡　* # (NT$ 60-65k)	M23-023	Deep enteroscopy. 確認檢查從上做/從下做 確認是否需全身麻醉：會診麻醉科，請 CR 輸入排程
EUS　* #	M23-051 M23-051B M23-120	Electronic endoscopic ultrasonography Miniprobe endoscopic ultrasound Endoscopy under sedation and analgesia
EUS-FNA　* # (NT$ 7.8k+ 13k+2k)	M23-052 M23-120	Endoscopic ultrasound guide fine needle aspiration Endoscopy under sedation and analgesia
大腸鏡　*	M23-013	Colonofiberscopy
乙狀結腸鏡　*	M23-012	Sigmoidoscopy, flexible
ERCP　* #	M23-004 X75-751 M23-120	ERCP E.R.C.P. Endoscopy under sedation and analgesia
腸胃道金屬支架 * # (NT$ 65k+ 10k)	M23-080 X75-981	Esophageal metal stent placement Fluoroscopy
減痛-上(NT$ 2k)	M23-120	Endoscopy under sedation and analgesia
減痛-下(NT$ 2.5k)	M23-121	Colonoscopy under sedation and analgesia
減痛-上&下(NT$ 3k)	M23-122T	Endoscopy& colonoscopy under sedation& analgesia

- Colon prepare：PEG or Bowklean (PEG 4000 1PC + fleet enema 2 PC PRN)
- 侵入性檢查後(如 TACE/RFA/ERCP/POEM)，請注意生命徵象及是否出血/疼痛

超音波室（分機 2225, 2235）

腹部超音波	M23-034 M23-032	Abdominal echo, liver, pancreas, spleen Echo, follow（半年內做過）
Drainage #	M23-043A/B	Drainage of cyst or abscess of liver/ subphrenic abscess
Aspiration #	M23-041	Echo guide aspiration
Biopsy # FNAC #	M23-042 M23-041A/B	Echo guide biopsy 肝臟生檢 Abdominal puncture 肝臟細針抽吸細胞學檢查
RFA # ①確認健保或自費 ②邀請醫師做檢查 ③聯絡檢查室時段	M23-046 ? M23-123	Radiofrequency ablation（ ? 依腫瘤大小選 A/B/C） Liver tumor radiofrequency ablation

- 引流後請注意引流
- 量、顏色與病患生命徵象

Management of Ascites

一、 腹水的原因

Causes of ascites

Cirrhosis	81 percent
Cancer	10 percent
Heart failure	3 percent
Tuberculosis	2 percent
Dialysis	1 percent
Pancreatic disease	1 percent
Other	2 percent

二、 形成腹水的病理機轉

- 白蛋白合成下降及血管擴張，使得 effective arterial volume 不足，導致 RAA and sympathetic nervous system 及 ADH 上昇，導致水分滯留。

三、 腹水的嚴重分級

1. grade I: mild, 只能在超音波下看得到
2. grade 2: moderate, 腹水平均分佈在腹內，病人感到輕微腹脹
3. grade 3: massive,大量腹水，病人感到十分腹脹

四、 什麼情況下要抽腹水

1. 新發生的腹水
2. 肝硬化病人剛住院時如有合併腹水
3. 肝硬化病人住院中病情惡化

- 抽腹水的位置可選擇左下腹或右下腹，但以左下腹優先，離 anterior superior iliac crest 往內三公分及往上三公分，此處腹壁較薄，記得避開表面的 collateral vein
- 抽腹水前不需要 routine 輸 plt or FFP 因為 risk 非常低

五、　抽腹水需要檢驗哪些項目

1. 抽出來第一步先看 Grossappearance:

- Yellow:清澈淡黃色為一般腹水顏色
- Bloody: 懷疑 TB or malignancy, 加驗 TB smear/culture 或是 cytology, 如果是 fresh blood 要懷疑 hemoperitoneum, 請注意病人 vital sign 並且告知病房 CR
- Milky: 懷疑 chylousascites, 要加驗 TG（>200 確診）
- Bile: 懷疑 bile duct leakage
- Fecal:懷疑 bowel perforation,加驗 CEA (>5ng/ml) and Alk-P (>240 U/L)

2. Routine 一定要驗的項目：

- Cell count and differential count: 如有 bloody ascites 要校正 WBC 及 PMN 的數目, corrected WBC= substrate 1 WBC/ 750 RBC, corrected PMN= substrate 1 PMN/ 250 PMN
- albumin: 要算 SAAG 用的, SAAG≥1.1=portal hypertension, SAAG<1.1=non-portal hypertension
- Total protein: 可用來區分 cirrhotic ascites(TP<2.5)及 HF related ascites(TP>2.5), 在沒有 SBP 的病人如果 TP<1 的話，未來發生 SBP 的風險是 TP>1 的十倍

3. Optional（當懷疑病人有感染的時候可以送檢）

- Gram’s stain, ascites culture(bedside 直接打入 culture bottle)
- Glucose, LDH: if glucose <50, LDH>225 要小心 secondary peritonitis
- Amylase:當懷疑 pancreatic disease 可以驗（amylase>100）

4. Unusual:

- AFB smear and culture: lym predominant, exudative
- Cytology: lym predominant, exudative
- TG: milky ascites, suspect chylous ascites, (TG>200)

5. Unhelpful

- PH
- Lactate
- Cholesterol
- Fibronectin
- Glycosaminoglycans

6. 其他檢驗

- 懷疑 HF related ascites,可抽血液 BNP (>364)

7. 真的找不到原因的腹水可考慮會診外科做 laparoscopic

六、 腹水的治療

1. 第一線

- 戒酒
- 限鹽：low “sodium” diet 2g=88mmol/d= “5g” salt diet
- 利尿劑使用：以 spironolactone 為主,起始劑量為 100mg(2pc BID),最大劑量為 400mg(8pc BID)可搭配 lasix 使用,起始劑量為 40mg(1pc QD),最大劑量為 160mg(4pc QD)
- 停止或謹慎使用 NSAID
- 評估肝臟移植

2. 第二線

- 停止或謹慎使用 ACEI/ARB, β-blocker
- 在血壓較低病人可考慮使用 midodrine
- serial therapeutic paracentesis, 可考慮 large volume paracentesis (>5L/times) + albumin infusion(6-8g/L)
- 評估肝臟移植
- Transjugular intrahepatic portosystemicshunt (TIPS):可考慮使用在 refractory ascites 的病人，可當作換肝前的 bridge,可有效控制腹水但是肝昏迷的發生率會上昇，重點是本院沒有在做！！

3. 第三線

- Peritoneovenous shunt:盛行於 1970 年代，但因為 high complication rate 而淪為時代的產物，現已無人在做。

七、 何謂 refractory ascites

1. 期間：利尿劑積極治療（spironolactone 400 mg/日與 furosemide 160 mg/日）至少一周，且病人接受限鹽飲食（<90 mmol/日或 5 g/日）
2. 無反應：平均體重，於四天中下降不到 0.8 公斤，且尿液鈉輸出量小於攝入量。
3. 腹水很快復發：最初處置開始後四週內復發第二到三級的腹水。
4. 利尿劑併發症：
    - 利尿劑引致之肝腦病變：無其他引致因素下發生的腦病變。

- 利尿劑引致之腎病變：在腹水治療有效的病人，血清 creatinine 值增加超過 100% 至大於 2 mg/dL
- 利尿劑引致之低血鈉：血清鈉離子降低大於 10 mmol/L 至 125 mmol/L 以下。
- 利尿劑引致之低血鉀或高血鉀：即便經過適當處置，血清鉀離子濃度仍小於 3 mmol/L 或大於 6 mmol/L。

Classification of ascites by the serum-to-ascites albumin gradient

High albumin gradient (SAAG ≥1.1 g/dL)
Cirrhosis
Alcoholic hepatitis
Heart failure
Massive hepatic metastases
Heart failure/constrictive pericarditis
Budd-Chiari syndrome
Portal vein thrombosis
Idiopathic portal fibrosis
Low albumin gradient (SAAG <1.1 g/dL)
Peritoneal carcinomatosis
Peritoneal tuberculosis
Pancreatitis
Serositis
Nephrotic syndrome

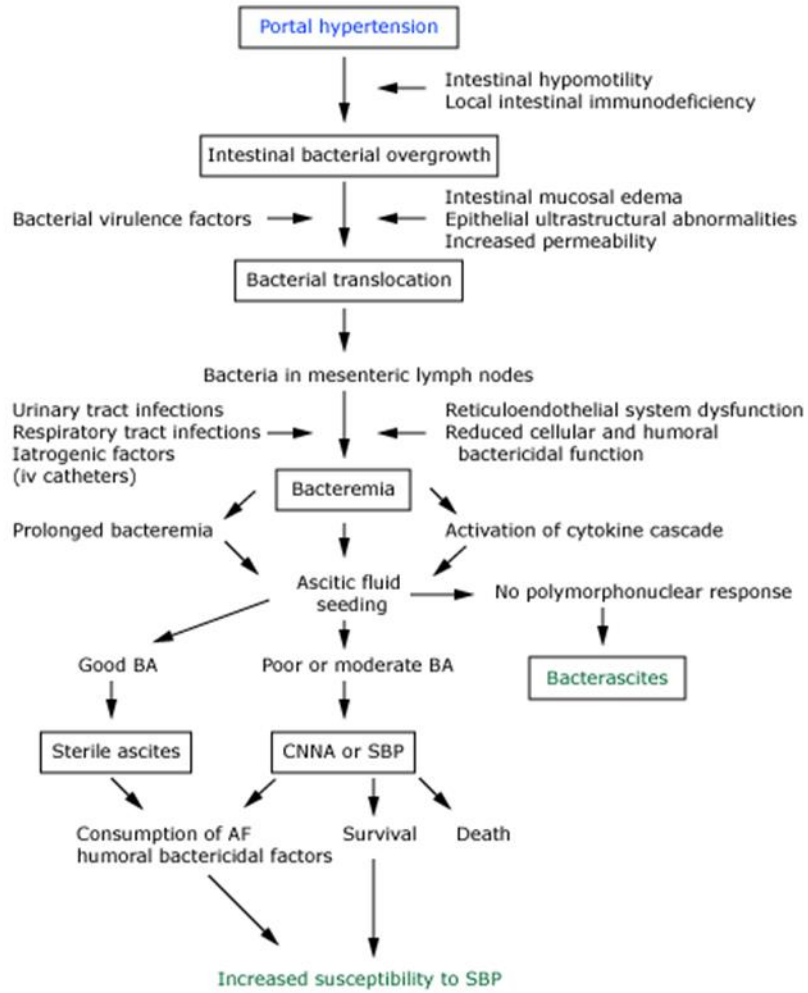
Spontaneous bacterial peritonitis

- Introduction

SBP is a very common in patients with cirrhosis and ascites. Prevalence of SBP in outpatients is 1.5–3.5%. When first described, its mortality exceeded 90%. And it could be reduced to approximately 20% with early diagnosis and treatment.

就跟 DM 有 DM foot 一樣, cirrhosis 的病人會發生 SBP, cirrhosis 的病人來發生 sepsis 一定要想到.

Mechanisms that may be involved in the pathogenesis of spontaneous bacterial peritonitis



BA: bactericidal activity; CNNA: culture-negative neutrocytic ascites; SBP: spontaneous bacterial peritonitis.

- Symptoms

- Local symptoms and/or signs of peritonitis:
  - Abdominal pain
  - Abdominal tenderness
  - Vomiting
  - Diarrhea
  - Ileus
- Signs of systemic inflammation:
  - Hyper or hypothermia, chills
  - Altered white blood cell count
  - Tachycardia
  - Tachypnea
- Worsening of liver function
- Hepatic encephalopathy
- Shock
- Renal failure
- Gastrointestinal bleeding

However, it is important to point out that SBP may be asymptomatic, particularly in outpatients.

一言以蔽之，肚子不舒服或是有 sepsis 的 signs, 就要想到 SBP.

- Diagnosis

Elevated ascitic fluid absolute polymorphonuclear leukocyte (PMN) count>250 cells/mm [0.25*109/L]) without an evident intra-abdominal, surgically treatable source of infection.

這邊要稍微說一下，這個指的是 Spontaneous bacterial peritonitis, 其他像是腹膜透析(CAPD)造成的 PD peritonitis, 並不適用使用這個診斷方式, 像是 PD peritonitis 只需要 PMN) count> 100 cells/mm.

- Diagnosis

  Abdominal paracentesis must be done before antibiotics applied. Empiric treatment without a sample for testing could not narrowing the spectrum of coverage. Due to even a single dose of an effective broad-spectrum drug causes the culture to produce no growth.

老生常談了, 給抗生素之前必留檢體.

- Treatment

If ascitic PMN counts greater than or equal to 250 cells/mm, empiric antibiotic therapy should be initiated.

Isolation of bacteria from fluid samples will be maximized if

- Cultured in blood culture bottles
- No prior antibiotic treatment
- No other explanation for an elevated PMN count
  - Hemorrhagic ascites, peritoneal carcinomatosis, pancreatitis, or peritoneal tuberculosis

Relatively broad-spectrum therapy is warranted in patients with suspected ascitic fluid infection until the results of susceptibility testing are available.

Cefotaxime, a third-generation cephalosporin.Shown to be superior to ampicillin plus tobramycin in a controlled trial. It covers 95% of the fl ora including the 3 most common isolates:

- Escherichia coli
- Klebsiella pneumoniae
- Streptococcal pneumoniae

我們使用的是相等的 ceftriaxone

參考我們 2015 年腹水培養的比率是 E.coli 第一, yeast like 可能是跟開腹術後或是 PD 比較有相關, 其他還有 enterococcus, K.P 以及 pseudomonas.

2015 ASCITES CULTURE IN CGMH

'E.coli'	303
'Yeast-like'	214
'Enterococcus faecalis'	146
'Kleb.pneumoniae'	130
'Ps.aeruginosa'	102
'B.fragilis'	79
'Enterococcus faecium (VRE)'	73
'Candida albicans'	70
'Enterococcus faecium'	67
'Chryseobacterium indologenes (Flavob.ind'	59
'Staph.epidermidis'	59
'Strepto.anginosus'	54
'Stenotrophomonas maltophilia (Xan.maltop'	50
'Staph.haemolyticus'	47
'B.thetaiotaomicron'	46
'Gm(+) no spore forming bacilli'	45
'Enterob.cloacae'	42
'Staph.aureus'	37
'Acinetobacter sp-MDR strain'	29
'Enterococcus avium'	28
'Gm(-) bacilli-glucose nonfermenting grou'	28
'Viridans streptococcus'	28

Intravenous Albumin Infusion in Addition

Mortality improved if albumin is applied additionally with antibiotics.

- Albumin should be given when
  - Serum creatinine is > 1 mg/dl or
  - Blood urea nitrogen > 30 mg/dl or
  - Total bilirubin > 4 mg/dl
- 1.5 g albumin per kg body weight within 6 hours and 1.0 g/kg on day 3
- Not necessary in patients not meet these criteria

- Follow-up Paracentesis

A follow-up ascitic fluid analysis is not needed in many patients with infected ascites.

Only needed in

- Nosocomial setting
- Presence of recent b-lactam antibiotic exposure
- Culture an atypical organism(s)
- Distinguishing Spontaneous From Secondary Bacterial Peritonitis

Secondary peritonitis can be divided into two subsets:

- With free perforation of a viscus
- With lobulated abscesses

Characteristic analysis in the setting of free perforation

- PMN count greater than or equal to 250 cells/mm
  - Usually many **thousands**
- **Multiple organisms** on gram' s stain and culture
  - Frequently including fungi and enterococcus
- At least two of the following criteria:
  - **Total protein** greater than 1g/dl
  - **LDH** greater than the upper limit of normal for serum
  - **Glucose**< 50 mg/dl
- 100% sensitivity but only 45% specificity in detecting perforation
- **Computerized tomographic scan** was diagnostic in **85%** of patients

In detecting **gut** perforation into ascitic fluid

- Ascitic fluid **carcinoembryonic antigen** greater than 5 ng/mL
- Ascitic fluid **alkaline phosphatase** greater than 240 units/L

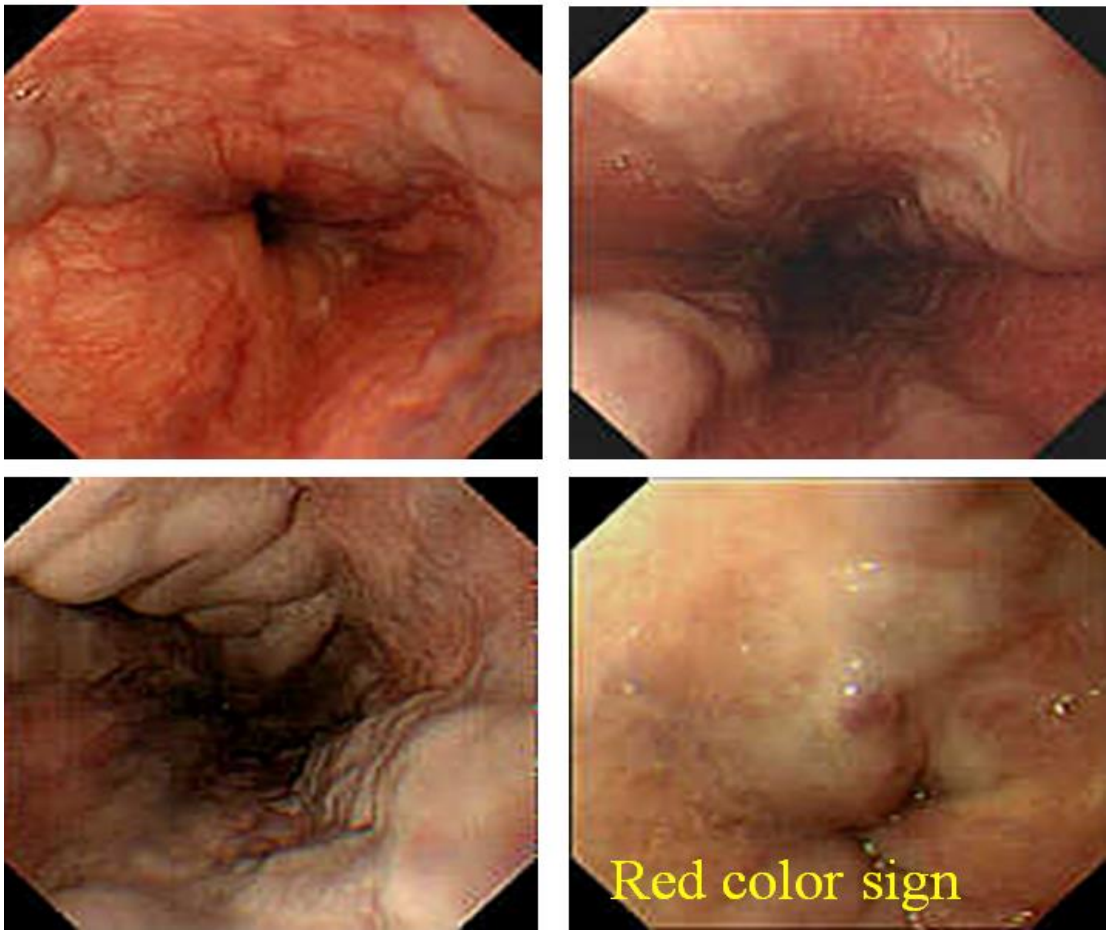
Sensitivity of 92% and specificity of 88%

- Patients who fulfill either set of criteria for gut perforation should undergo emergent **computed tomographic scan**

Management of Varices in patients with liver cirrhosis

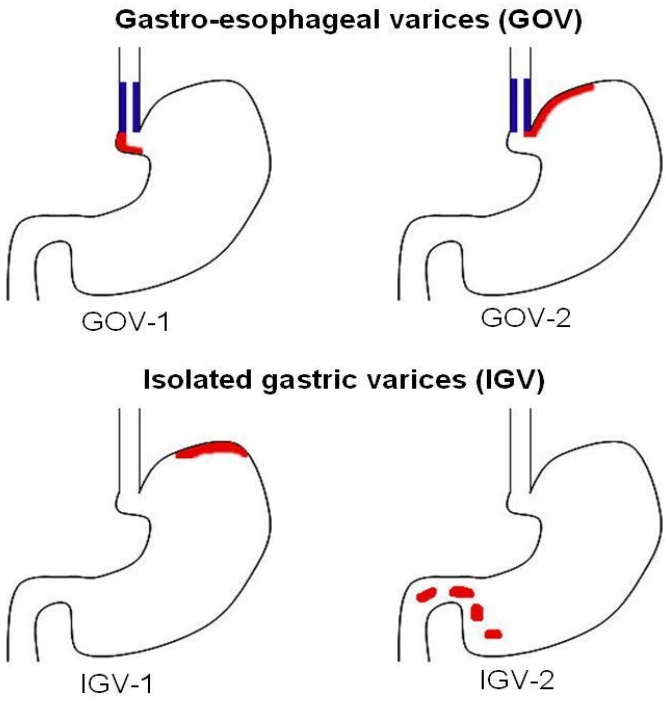
Definition

Esophageal varices(EV): F1(左上), F2(右上), F3(左下)



根據Japanese Society for Portal Hypertension的分類及定義
Form 1 (F1): Straight small-caliber varices
Form 2 (F2): Moderately enlarged, beady varices
Form 3 (F3): Markedly enlarged, nodular or tumor-shaped varices
Red Color Sign:
Red wale marking (linear capillaries)
Cherry red spot (red spots)
Hematocystic spot

Gastric varices



Risk factors

- Portal pressure:
  - Clinically significant portal hypertension: HVBG >10mmHg
  - HVBG > 12mmHg: variceal rupture
- Variceal size: > 5 mm
  - Endoscopic feature: red wale sign (red color sign)
- Severity of liver diseases

Survey and follow-up:

1. EGD in all patients with cirrhosis at the time of diagnosis (1a, A). There is no indication to repeat endoscopy in patients receiving non-cardioselective β blockers (NSBB).
   i. No varices: EGD at 2–3-year intervals (2a, B).
   ii. Grade I varices(F1): EGD at yearly intervals (2a, B).
   iii. If there is clear evidence of disease progression, the intervals can be modified by the clinician.
2. EGD at time of decompensation (2a, B).

Pre-primary prophylaxis (prevention of varices formation)

1. Treat underlying liver diseases
2. Beta-blockers has no indication to prevent varices formation.

Primary prevention

A. Esophageal varices

1. Small varices(F1) with red wale signs or Child C class : NSBB (1b, A)
2. Medium-large varices(F2, F3): EVL or NSBB (1a, A).
3. Dose: Propranolol: 40 mg bid. Dose titrated to maximum tolerated or once heart rate (HR) of 50–55 bpm (max dose: 320 mg) (1a, A).

   Alternatives:

   - Nadolol: 40 mg qd. Dose titrated to maximum tolerated or once HR of 50–55 bpm (max dose: 240 mg) (1a, A).
   - Carvedilol: 6.25 mg qd to increase to maintenance of 12.5 mg after a week if tolerated or once HR of <50–55 bpm is reached (1a, A).
   - Spontaneous bacterial peritonitis, renal impairment and hypotension => Discontinue NSBB (2b, B).
4. Treatments not recommended as primary prevention:
   - × NSBB in combination with Isosorbide-5-Mononitrate (ISMN), spironolactone, or EVL
   - × Proton pump inhibitors unless otherwise required for peptic ulcer disease (1b, B).
   - × Isosorbide mononitrate monotherapy
   - × Shunt surgery or transjugular intrahepatic portosystemic shunt (TIPS) or Sclerotherapy (1a, A)

B. Gastric varices: NSBB (2a, B).

Management of acute active bleeding

A. **Suggestions for resuscitation and initial management**

1. A restrictive transfusion policy aiming for a hemoglobin of 7–8 g/dL is suggested in hemodynamically stable patients

(1b, B).

- Patients without actively bleeding and unstable hemodynamically stable: Do not offer platelet transfusion (5, D).
- Platelet transfusion: patients with actively bleeding and platelet count of $<50\times10^9$/L (5, D).
- FFP: patients who have either a fibrinogen level of <1 g/L (5, D), or a PT (INR) or aPTT >1.5 times normal (5, D).
- Prothrombin complex concentrate: patients who are taking warfarin and actively bleeding (5, D).

2. Warfarin should be stopped as local warfarin protocols (5, D).
3. Recombinant factor VIIa: insufficient evidence(1b, B).

B. Timing of upper gastrointestinal endoscopy

1. Unstable patients: EGD immediately after resuscitation (5, A).
2. EGD within 12 h of admission to all other patients (5, D).

C. **Control of bleeding**

Esophageal varices

1. Antibiotics(1a, A) with Ciprofloxacin or Ceftriaxone IVF 3-7 days
2. Vasoconstrictors such as terlipressin or somatostatin as soon if variceal bleeding is suspected and should be continued until hemostasis is achieved or for up to 5 days.
   - Alternative: Octreotide if no terlipressin or somatostatin(1a, A).
   - Dose: terlipressin:1~2gm q6h ~ q4h for max: 5 days
     Octreotide: 0.1 mg iv every eight hours for two days. 100 mcg once then 25 mcg/h for 24 h then 100 mcg sc tid on days two-seven. Octreotide 25 mcg/h infusion for 12 h then 100 mcg sc at 12 h and 18h
3. Endoscopic variceal ligation (1a, A).

4. Early covered TIPS after satisfactory hemostasis with the methods above(<72 h after index variceal bleed) can be considered in selected patients with Child’ s B cirrhosis and active bleeding or Child’ s C cirrhosis with Child’ s score <14 (1b, B).
5. No PPI unless otherwise required for peptic ulcer disease (1b, B).
6. Failure to control active bleeding
   - Sengstaken– Blakemore tube (1b, B).
   - TIPS with PTFE-covered stents or surgery (1b, B).

Gastric varices

1. Gastro-oesophageal varices (GOV)-1: treat as for EV (2b, B).
2. GOV-2 and isolated gastric varices (IGV):
   - Initial endoscopic therapy with cyanoacrylate injection
   - TIPS can be considered (3a, B).
3. If control of bleeding fails:
   - Balloon tamponade first for GOV and IGV-1 until definitive treatment is undertaken (2b, B).
   - The first-line definite treatment: Salvage TIPS (3a, B).
   - Balloon-occluded retrograde transvenous obliteration (B-RTO) or surgical shunting can be considered if TIPS is not possible (eg, portal vein thrombosis present) (3a, B).

Secondary prevention

A. Esophageal varices

1. NSBB (propranolol or nadolol)+EVL combination therapy (1a, A).
2. **Alternative:** NSBB with ISMN or EVL monotherapy (1a, B)
   - NSBB: propranolol, nadolol, or carvedilol (1b,B).
   - If NSBB alone are used, there is no need to undertake further endoscopy unless clinically indicated (1a, A).
   - EVL alone if there are contraindications or intolerance to combined use with NSBB (1a, A)

3. Varices are banded at 2–4-weekly intervals until eradication (1b, B).
   - After successful eradication of the varices: EGD at 3 months, then 6 monthly thereafter. (1b, B).
     ✓ Any recurrent varices =>further EVL until eradication (1b, B).
4. No PPI unless otherwise required for peptic disease (1b,B).
5. TIPS with PTFE covered stents is used for patients who re-bleed despite treatment as above description (1a, A).
6. Where TIPS is not feasible in Child' s A and B patients, shunt surgery can be used(1b, B).

B. Gastric varices

1. GOV-1: variceal band ligation (2b, B).
2. GOV-2 and IGV: endoscopic surveillance with cyanoacrylate injection (2b, B)
3. NSBB can be considered in certain circumstances (1b, B).
4. TIPS if patients re-bleed despite cyanoacrylate injection. TIPS can also be considered in other selected patients (eg, those with large or multiple gastric varices) (1b, B).
5. Shunt surgery may be used in selected patients with well-compensated cirrhosis (3c, B).
6. Splenic vein thrombosis or left-sided portal hypertension => Splenectomy or splenic artery embolization(4, C).

Hepatic encephalopathy

- Introduction

Hepatic encephalopathy is a brain dysfunction caused by liver insuffi ciency and/or PSS ranging from subclinical alterations to coma.

Overt hepatic encephalopathy means the grade of hepatic encephalopathy is more severe. It is defined as onset of **disorientation** or asterixis by the International Society for Hepatic Encephalopathy and

Table 2. WHC and clinical description.

WHC including MHE	ISHEN	Description	Suggested operative criteria	Comment
Unimpaired		No encephalopathy at all, no history of HE	Tested and proved to be normal	
Minimal	**Covert**	Psychometric or neuropsychological alterations of tests exploring psychomotor speed/executive functions or neurophysiological alterations without clinical evidence of mental change.	Abnormal results of established psychometric or neuropsychological tests without clinical manifestations	No universal criteria for diagnosis. Local standards and expertise required
Grade I		• Trivial lack of awareness • Euphoria or anxiety • Shortened attention span • Impairment of addition or subtraction • Altered sleep rhythm	Despite oriented in time and space (see below), the patient appears to have some cognitive/behavioural decay with respect to his/her standard on clinical examination, or to the caregivers	Clinical findings usually not reproducible
Grade II	**Overt**	• Lethargy or apathy • Disorientation for time • Obvious personality change • Inappropriate behavior • Dyspraxia • Asterixis	Disoriented for time (at least three of the followings are wrong: day of the month, day of the week, month, season or year) ± the other mentioned symptoms	Clinical findings variable but reproducible to some extent
Grade III		• Somnolence to semi-stupor • Responsive to stimuli • Confused • Gross disorientation • Bizarre behavior	Disoriented also for space (at least three of the following wrongly reported: country, state [or region], city or place) ± the other mentioned symptoms	Clinical findings reproducible to some extent
Grade IV		Coma	Does not respond even to pain stimuli	Comatose state usually reproducible

All conditions are required to be related to liver insufficiency and/or PSS.

Nitrogen Metabolism consensus.

- Epidemiology

Prevalence of overt HE at the time of diagnosis of cirrhosis

- 10%-14% in general
- 16%-21% in decompensated cirrhosis
- 10%-50% in patients with trans-jugular intrahepatic portosystemic shunt (TIPS)

- Epidemiology

The first bout of overt HE occurs within 5 years in 5%–25% of population after cirrhosis diagnosis. Risk factors include complications to cirrhosis (MHE or CHE, infections, VB, or ascites), probably diabetes, hepatitis C.

If there is previous of overt HE, there is 40% cumulative risk at 1 year for recurring overt HE.

And if there is once recurrent overt HE, there is 40% cumulative risk in 6 months for another recurrence.

- Classification

  This classification is according to the underlying disease, time course and precipitating factors.

Underlying disease

- Type A

  Resulting from acute liver failure
- Type B

  Resulting predominantly from PSS
- Type C

  Resulting from cirrhosis

Time course

- Episodic HE
- Recurrent HE

  Occur with a time interval of 6 months or less
- Persistent HE

  A pattern of behavioral alterations that are always present and interspersed with relapses of overt HE

Precipitating factors

- Nonprecipitated
- Precipitated

Episodic	Recurrent
Infections*	Electrolyte disorder
GI bleeding	Infections
Diuretic overdose	Unidentified
Electrolyte disorder	Constipation
Constipation	Diuretic overdose
Unidentified	GI bleeding

- Classification

Type	Grade		Time Course	Spontaneous or Precipitated
A	MHE	Covert	Episodic	Spontaneous
	1		Recurrent	
B	2	Overt		Precipitated (specify)
	3		Persistent	
C	4			

Overt HE or acute confusional state
- Diabetic (hypoglycemia, ketoacidosis, hyperosmolar, lactate acidosis)
- Alcohol (intoxication, withdrawal, Wernicke)
- Drugs (benzodiazepines, neuroleptics, opioids)
- Neuroinfections
- Electrolyte disorders (hyponatremia and hypercalcemia)
- Nonconvulsive epilepsy
- Psychiatric disorders
- Intracranial bleeding and stroke
- Severe medical stress (organ failure and inflammation)

Other presentations
- Dementia (primary and secondary)
- Brain lesions (traumatic, neoplasms, normal pressure hydrocephalus)
- Obstructive sleep apnea

Hyponatremia and sepsis can both produce encephalopathy *per se* and precipitate HE by interactions with the pathophysiological mechanisms. In end-stage liver disease, uremic encephalopathy and HE may overlap.

- Differential Diagnoses
- Recommendations according to the guideline

1. Hepatic encephalopathy should be classified according to the type of underlying disease, severity of manifestations, time course, and precipitating factors (GRADE III, A, 1).
2. A diagnostic workup is required, considering other disorders that can alter brain function and mimic HE (GRADE II-2, A, 1).

- Diagnosis and Testing

Clinical scales are used to analyze its severity. Specific quantitative tests are only needed in study settings. There is no need for quantitative tests in establish diagnosis for hepatic encephalopathy. Generally, we use Glasgow Coma Scale for evaluate conscious disturbance.

Testing for minimal HE (MHE) and convert HE (CHE) including

- Portosystemic encephalopathy syndrome test
- The Critical Flicker Frequency test
- The Continuous Reaction Time test
- The Inhibitory Control Test
- The Stroop test
- The SCAN Test
- Electroencephalography examination

- Ammonia

High blood-ammonia levels alone do not add any diagnostic, staging, or prognostic value for patient with suspect hepatic encephalopathy. If in case an ammonia level is checked in a patient with overt HE and the blood-ammonia level is normal, the diagnosis of HE is in question.

亦即, HE 的診斷來自於意識障礙的臨床評估, 以 clinical scale 的評估為主, 包含 Glasgow Coma Scale. 血液中 ammonia level 單獨無法構成 HE 的診斷. 但是懷疑 HE 的病人如果 ammonia 不高, 則意識障礙的原因可能並不是 hepatic encephalopathy.

For ammonia-lowering drugs, repeated measurements of ammonia may be helpful to test the efficacy.

- Brain Scans

Do not contribute diagnostic or grading information for patient with HE. However, risk of intracerebral hemorrhage is at least 5-fold increased in patient with impaired liver function. Bain scan is for differential diagnosis to exclude ICH. Due to symptoms may be indistinguishable between HE and ICH.

- Recommendations according to the guideline

3. Hepatic encephalopathy should be treated as a continuum ranging from unimpaired cognitive function with intact consciousness through

coma(GRADE III, A, 1).
4. The diagnosis of HE is through exclusion of other causes of brain dysfunction (GRADE II-2, A, 1).
5. Hepatic encephalopathy should be divided into various stages of severity, reflecting the degree of self-sufficiency and the need for care (GRADE III, B, 1).
6. Overt hepatic encephalopathy is diagnosed by clinical criteria and can be graded according the WHC and the GCS (GRADE II-2, B, 1).
7. The diagnosis and grading of MHE and CHE can be made using several neurophysiological and psychometric tests that should be performed by experienced examiners (GRADE II-2, B, 1).
8. Testing for MHE and CHE could be used in patients who would most benefit from testing, such as those with impaired quality of life or implication on employment or public safety (GRADE III, B, 2).
9. Increased blood ammonia alone does not add any diagnostic, staging, or prognostic value for HE in patients with CLD. A normal value calls for diagnostic reevaluation (GRADE II-3, A, 1).

- Treatment

Only overt HE is routinely treated. And for patients with higher grades of HE, protect airway and ICU care is needed.

- Recommendations according to the guideline

10. An episode of overt HE (whether spontaneous or precipitated) should be actively treated (GRADE II-2, A, 1).

11. Secondary prophylaxis after an episode for overt HE is recommended (GRADE I, A, 1).

12. Primary prophylaxis for prevention of episodes of overt HE is not required, except in patients with cirrhosis with a known high risk to develop HE (GRADE II-3, C, 2).

13. Recurrent intractable overt HE, together with liver failure, is an indication for liver transplant (GRADE I).
14. Initiation of care for patients with altered consciousness
15. Alternative causes of altered mental status should be sought and treated.

16. Identification of precipitating factors and their correction
17. Commencement of empirical HE treatment

- Therapy for Episodes of overt HE
  - Nonabsorbable Disaccharides
  - Rifaximin
  - Other Therapies
    - BCAAs
    - Metabolic Ammonia Scavengers
    - L-ornithine L-aspartate (LOLA)
    - Probiotics
    - Glutaminase Inhibitors
    - Neomycin
    - Metronidazole
    - Flumazenil
    - Laxatives
    - Albumin

- Lactulose (Nonabsorbable Disaccharides)

The dosage is initiated with25 mL Q12H and the dosage will be increased until there is at least two times of soft or loose bowel movements per day. Dosing titrated to maintain two to three bowel movements per day

以病房來說，經常會有的事情就是病人說大便變得水水的，於是乎 lactulose 就會被 hold 或是直接 DC，但是直接參照 guideline 來說兩次以上 soft or loose bowel movements 這個是治療本來就設定的目標，並不是 complication，只要不是拉到超離譜，並不需要直接停用或是使用止瀉藥物，至於達到目標後如何減量，如前段章節所述，可以參考血液中 ammonia 的 level，並且維持一天至少 2 到 3 次的大便.

There is a danger for overuse of lactulose

- Aspiration
- Dehydration
- Hypernatremia
- Severe perianal skin irritation
- Over-use can even precipitate HE

- Rifaximin

  口服抗生素, 不會經由腸胃道吸收到 systemic circulation, 簡單的說就是因為 ammonia 的製造一份是來自於腸胃道的細菌所製造, 再從腸胃道吸收到 systemic circulation, 因此藉由口服抗生素來減少腸道細菌的數量, 來進一步減少腸道 ammonia 的製造, 最後目的是達成血液中 ammonia level 下降.

  目前此藥需要專案進口, 在病房是使用 neomycin 作為替代.

- Branched chain amino acids

  口服的才有效果, 目前如果要使用的話需要病患自己出去外面買.



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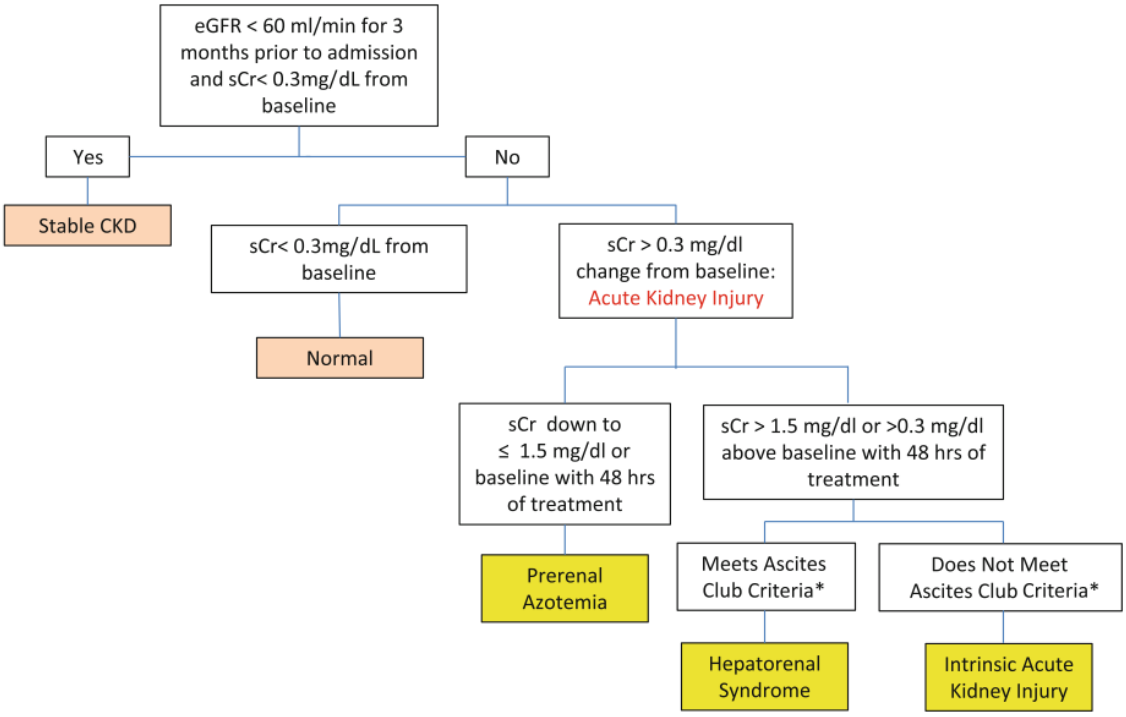
- L-ornithine L-aspartate

  目前台灣應該是沒有這類產品.

Hepatorenal syndrome

- Diagnosis
  - Cirrhosis with ascites
  - Creatinine > 1.5 mg/dl
  - No improvement of serum creatinine , after at least two days with diuretic withdrawal and volume expansion with albumin infusion in 1 g/kg of body weight per day up to a maximum of 100 g/d
  - Absence of shock
  - No nephrotoxic drugs
  - Absence of parenchymal kidney disease, including
    - Proteinuria >500 mg/day
    - Microhematuria
    - Abnormal renal ultrasonography

Hepatorenal syndrome 的診斷條件相對來說比較嚴格，畢竟 liver cirrhosis 的病人本身 complication 就多，感染之類的問題可能就可以解釋病人的 AKI，所以要診斷 Hepatorenal syndrome 單獨發生就必須要滿足上列的條件.



- Classification
  - Type I
    - Rapidly progressive reduction in renal function
    - Serum creatinine to a level > 2.5 mg/dl

- 50% reduction of the initial 24 hours creatinine clearance
- < 20 ml per minute in less than 2 weeks

- Type II
  - Not rapidly progressive course
  - Commonly associated with death in patients who not die of other complications of cirrhosis

- Prevention

Albumin, Pentoxifylline (In patient eGFR 41 and 80 mL/min)

- Treatment

Hemodialysis

Hemodialysis is frequently used to control azotemia and maintain electrolyte balance before liver transplantation.Hypotension during dialysis is a common problem. However, without transplantation survival is dismal.A more recent study reports that eight of 30 patients with HRS survived 30 days with use of hemodialysis or continuous venovenous hemodialysis in the intensive care unit setting.

Continuous venovenous hemofiltration/hemodialysis causes less hypotension but requires the continuous involvement of a dialysis nurse.

作為取代性治療，洗腎當然還是可以解決問題的方法之一啦.

Pharmacological Treatment

Vasoconstrictors

Terlipressin: 1 mg/4–6 hours intravenously. The dose is increased up to a maximum of 2 mg/4–6 hours after 3 days if there is no respons to therapy (defined as a 25% or greater reduction in serum creatinine compared to pretreatment levels). A response to therapy is defined as a decrease in serum creatinine levels below 1.5 mg/dL (133 µmol/L). The treatment is usually applied for 5 to 15 days.

Midodrine and octreotide: 7.5 mg of midodrine orally 3 times daily (increased to 12.5 mg 3 times daily if needed) and 100 µg of octreotid subcutaneously 3 times daily (increased to 200 µg 3 times daily if needed). The duration of treatment depends on the effects on serum creatinine.

Norepinephrine: 0.5–3 mg/hour as a continuous intravenous infusion aimed at increasing the mean arterial pressure by 10 mm Hg. The treatment is continued until the serum creatinine level decreases below 1.5 mg/dL.

Albumin Administration

Concomitant administration of albumin and vasoconstrictor drugs (1 g/kg of body weight on day 1 followed by 20–40 g/day).

Solà, Elsa, Mónica Guevara, and Pere Ginès. "Current treatment strategies for hepatorenal syndrome." *Clinical Liver Disease* 2.3 (2013): 136-139.

Hepatorenalsyndrome 的機制還是沒有一個很詳細的方式可以做出完整的解釋，不過大致上我是認為可以想像成血管中血液的分布與分配，由於 portal hypertension 以及 ascites formation 的關係而造成某些改變，因此腎臟的灌流下降，所以治療上使用 vasoconstrictor，在大部分血管收縮的同時，由於腎臟的 afferent arteriole 較不受 vasoconstrictor 影響，而達成血管中血液的分配再改變，最後達成腎臟血液灌流的增加.

- Recommendations according to the guideline

1. Urinary biomarkers such as neutrophil gelatinase associated lipocalin may assist in the differential diagnosis of azotemia in patients with cirrhosis.(Class IIa, Level B)
2. Albumin infusion plus administration of vasoactive drugs such as octreotide and midodrine should be considered in the treatment of type I hepatorenal syndrome. (Class IIa, Level B)
3. Albumin infusion plus administration of norepinephrine should also be considered in the treatment of type I hepatorenal syndrome, when the patient is in the intensive care unit. (Class IIa, Level A)
4. Patients with cirrhosis, ascites, and type I or type II hepatorenal syndrome should have an expedited referral for liver transplantation. (Class I, Level B)

Update of acute hepatitis and acute liver failure

Abnormal liver function tests 肝功能異常

一. 肝功能異常的判讀必須熟記各個 serum marker 的來源及半衰期，粗略來分肝功能異常至少能分成 Hepatocellular pattern 及 Cholestatic pattern 兩種 (詳見 Table1 及 Table2)：

TABLE 1. STANDARD LIVER TESTS, THEIR SOURCES OF ORIGIN, AND ABNORMALITIES

Parameter	Origin	Associated Disease
AST	Liver, skeletal muscle, cardiac muscle, red blood cells, brain, pancreas, lungs	Hepatocellular injury of any cause, myopathies, myocardial infarct, hemolysis
ALT	Liver, kidneys, skeletal muscle	Hepatocellular injury of any cause, myopathies
AP	Liver, bone, placenta, kidneys, intestines	Cholestatic liver disease; sarcoidosis; pregnancy; lymphoma; bone, kidney, and intestinal diseases
γ-Glutamyl transferase	Biliary epithelial cells, kidneys, pancreas, prostate	Biliary or pancreatic disease, myocardial infarct, renal diseases, chronic lung disease, diabetes
Conjugated bilirubin	Hemolysis, insufficient excretion from the liver	Severe liver injury from any cause Rotor syndrome, Dubin-Johnson syndrome
Unconjugated bilirubin	Hemolysis	Hemolysis, Gilbert syndrome, Crigler-Najjar syndrome
Albumin	Produced in hepatocytes	Low in nephrotic syndrome, malnutrition, protein-losing enteropathy
Prothrombin time	Clotting factors produced in hepatocytes	Prolonged in liver disease, vitamin K deficiency, fat malabsorption, pancreatic insufficiency

*AP:Alk-P

TABLE 2. ELEVATION OF LIVER CHEMISTRIES WITH LIVER DISEASES

Test	Hepatocellular	Cholestatic	Half-life ($t_{1/2}$)
AST	+++	N/+	17 hours
ALT	+++	N/+	47 hours
AP	Normal/mild	++++	7 days
gGT	++/+++	++++	26 days (for abstinence)
Total bilirubin	N/++	N/+++	Depends on albumin binding
Albumin	++ (chronic)	N	20 days
Prothrombin time	++	N	

Reference：CLINICAL LIVER DISEASE, VOL 8, NO 1, JULY 2016

Acute hepatitis

- **Deffinition:Diffuse liver inflammation lasting less than 6 months**

1. Acute Viral Hepatitis

	Hepatitis A HAV Infective hepatitis	Hepatitis B HBV serum hepatitis	Hepatitis D HDV	Hepatitis C HCV Post transfusion hepatitis	Hepatitis E HEV Epidemic/ Enteral
virus	27 nm RNA	42 nm DNA hepa virus	35 nm Incomplete RNA+HBsAg	30 – 60 nm RNA flavivirus	32 nm RNA
transmission	Feco-oral	■ Paraenteral and post transfusion ■ Sexual low risk 1-5 % ■ Intrauterine low risk <5%			Feco-oral
Incubation period	2 – 6 weeks	2 – 6 months	2 – 6 months	2 – 6 months	2 – 6 weeks
Chronicity & liver cancer	no	Yes	Yes	Yes	No
Immunization-passive	Non specific Ig	Specific Ig	IgM		Non specific Ig

1. **Pathogenesis:**
    1. **Initial viremia, with inflammation of gastrointestinal tract mucosa.**
    2. **Intrahepatic localization**
       A- diffuse centrilobular necrosis, with cellular infiltration around portal tracts
       B- intrahepatic cholestasis due to cellular edema & inspissation of bile
2. **Clinical picture**
    1. **Prodromal stage: 前驅期 3 – 9 days: Sudden onset of influenza like picture**
    2. **Icteric stage: 黃疸期 2-4 wk，若黃疸出現症狀已改善則可能這次 episode 能平安度過，若症狀仍未改善則須密切觀察或住院**

3. **Convalescence stage: 恢復期 >4weeks: Signs & symptoms gradually disappear. Jaundice may persist for some times due to affinity of bile pigment to elastic tissue**

3. **Lab test：**
    1. 密切追蹤肝功能 AST, ALT, Bil(T), PT(INR), Alpha-fetoprotein(A-FP)
    2. Seromarker 查 etiology：Anti-HAV IgM, HbsAg, Anti-HBc IgM, Anti-HCV,通報疾管局急性 E 肝

Hepatitis B virus: Clinical manifestations and natural history B 型肝炎

一. B 型肝炎需區分為急性 B 肝(**acute HBV infection**)及慢性 B 肝急性發作(**HBV reactivation**)

二. Serologic markers ：

- **Hepatitis B surface antigen (HBsAg)**：B 肝感染最重要證據，hallmark of HBV infection
- **Anti-HBs Ab**：具保護力之抗體，出現於感染後恢復及施打 B 肝疫苗後
- **Hepatitis B core antigen (HBcAg)**：為一重要之肝炎感染或是曾經感染證據
- **IgM anti-HBc**：為急性 B 肝(acute HBV infection)之重要指標
- **Hepatitis B e antigen (HBeAg)**：通常代表 B 肝之活躍程度(marker of HBV replication and infectivity)，通常 HBeAg 陽性患者有較高之病毒量(DNA)
- **Anti-HBe Ab**：e-seroconversion 越早之病患預後越佳

三. HBV 自然史

- B 肝各時期詳見詳見國際級大師廖運範/朱嘉明醫師下圖：

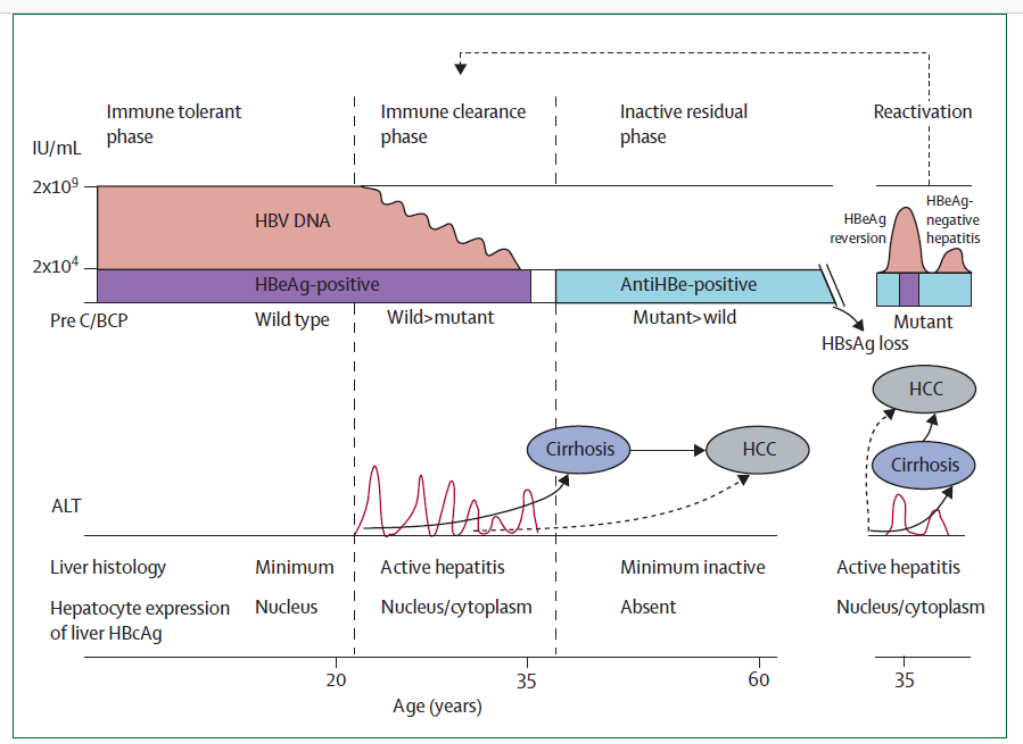


Figure 2: **Natural course of chronic hepatitis B virus (HBV) infection acquired perinatally and during infancy**

Reference：Yun-Fan Liaw, Chia-Ming Chu. Lancet 2009

Serology for HBV:

antigen	Significance	Corresponding Ab	significance
HBsAg	Appear after 6 week	Anti HBs	Appear after 3 m
	Acute infection, remain for 3 ms		Reflecting recovery & immunity
	Chronic infection if >6 ms		
HBcAg	Detected on Liver Biopsy only (not serum)	Anti HBc	Appear after 2 m
			Reflecting infection history
HBeAg	Reflect ongoing viral replication (chronicity)	Anti HBe	Appear after 2.5 m
			Non replicating virus
HBV DNA	Most sensitive indication for viral replication & chronicity		

Serology for HAV, HDV, HEV:

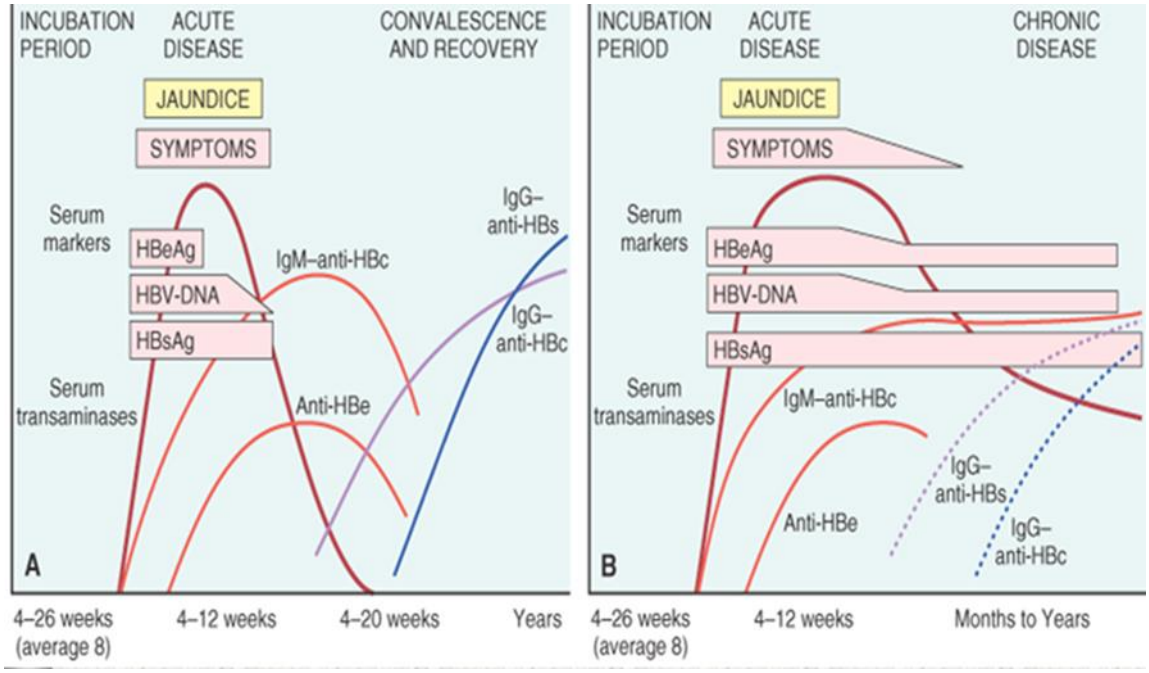
	Acute stage	Chronic stage	others
Hepatitis A	Anti HAV IgM	Anti HAV IgG	Fecal HAV
Hepatitis D	Anti HDV IgM	Anti HDV IgG	
Hepatitis E	Anti HEV IgM	Anti HEV IgG	

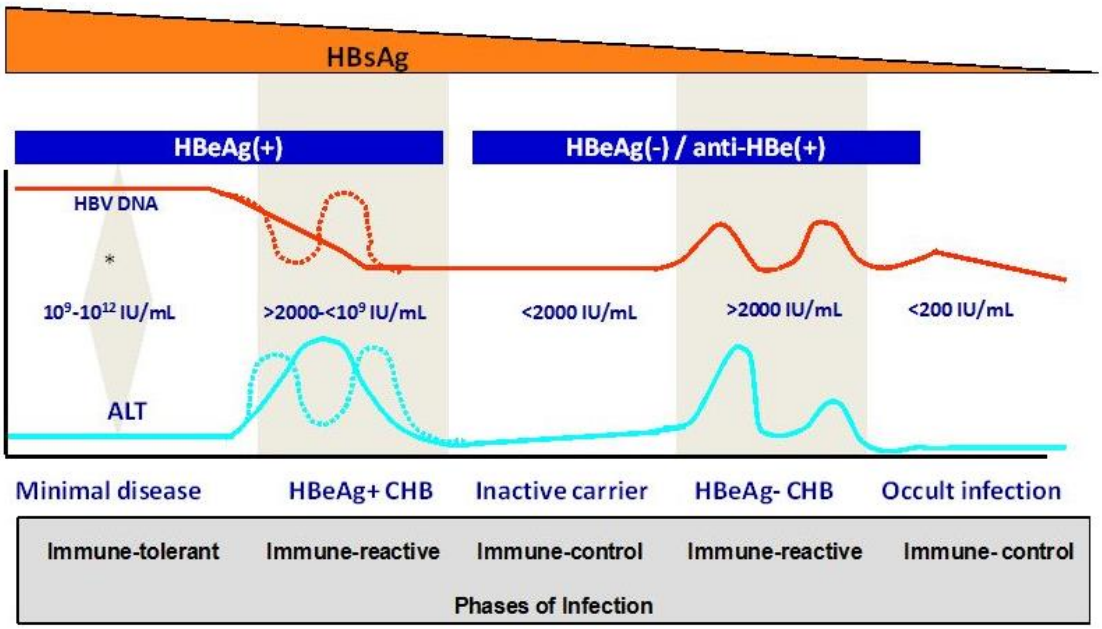
Serology for HCV

Marker	technique	time	significance
HCV Ab	2nd& 3rd generation ELISA or RIBA test	After 3 – 6 ms of infection	Exposure to infection not immunity
HCV RNA	PCR – quantitative PCR is more accurate & assess interferon titer	After 1 – 2 weeks	Indicate active virus

Hepatitis B

World Gastroenterology Organisation Global Guideline 2015





Acute hepatitis flare up in chronic hepatitis B

Hepatol 2014;11(4):209–219

Table 10 Causes of acute hepatitis flares of hepatitis in chronic hepatitis B virus infected patients

Spontaneous reactivation of hepatitis B virus replication
Due to immunosuppressive medications: cancer chemotherapy, antirejection drugs, corticosteroids
Cessation of anti-viral agents
Emergence of drug resistance
Due to antiviral therapy: interferon, corticosteroid withdrawal
Due to superimposed infections with other hepatotropic viruses: hepatitis A/E virus, hepatitis C virus, hepatitis delta virus
Caused by interaction with HIV infection: reactivated hepatitis, effect of immune reconstitution therapy
Other hepatotropic insults: drugs, alcohol

Management of hepatitis B

APASL 2016 guideline

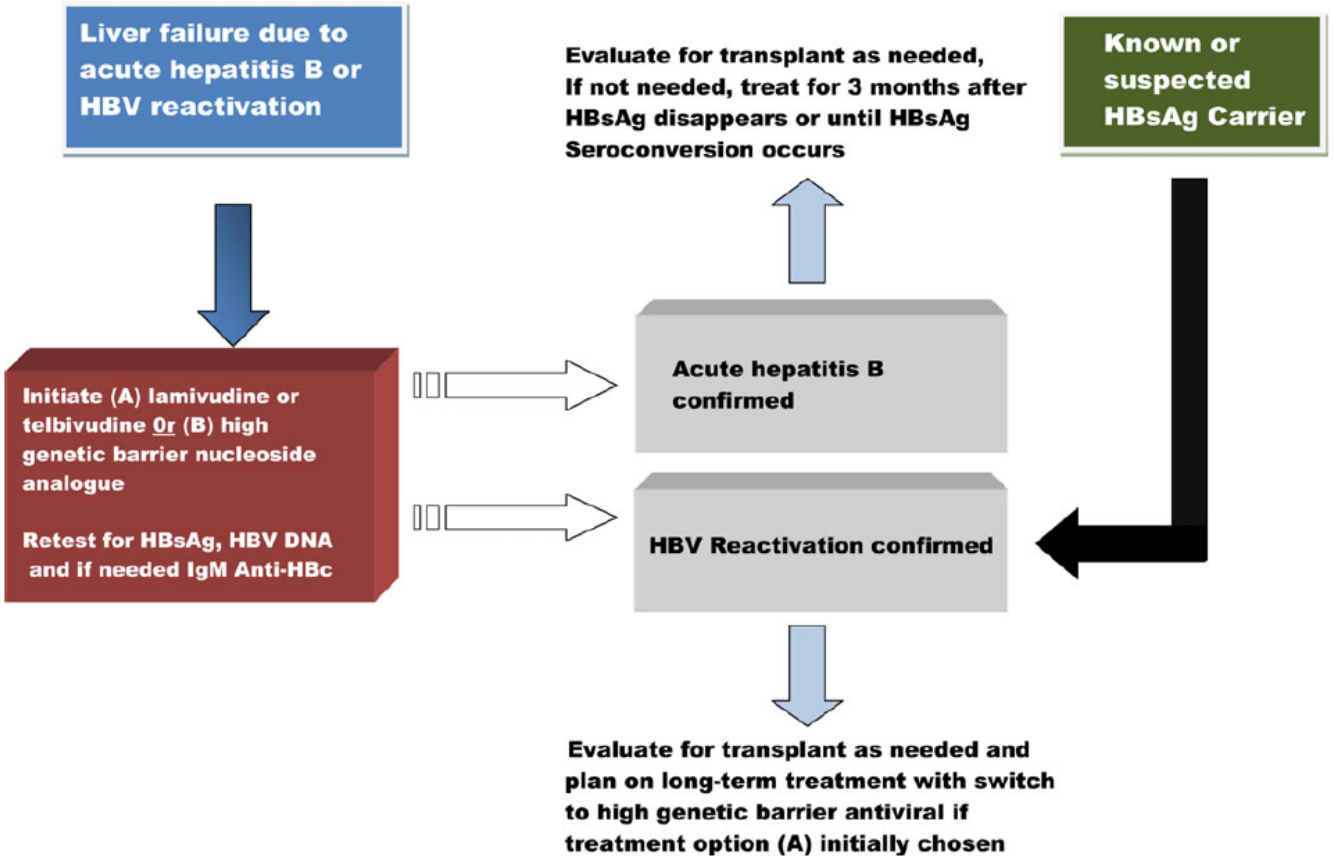


TABLE 1: Indications for Prompt or Urgent Treatment of Hepatitis B

	Major Supportive Data	Key Indications for Urgent Treatment	Preferred Treatment
Cirrhosis*			
Decompensated	Open label; multiple large case series	Clinical stabilization; minimize risk for recurrence posttransplantation	Entecavir (0.5 mg) or tenofovir (300 mg)
Borderline	Undefined	Forestall disease progression; avoidance of transplantation	As above
Well compensated	Randomized, controlled trials	As above	As above
Acute liver failure			
HBV reactivation	Open label with comparison to historical controls	Minimize further liver injury; reduce risk for recurrence after transplantation if needed	Entecavir (0.5 mg) or tenofovir (300 mg)†,‡
Severe acute hepatitis	As above, but smaller case series	Minimize further liver injury and enhance full recovery	Consider lamivudine or telbivudine§

*It has been the author's practice to use maintenance antiviral therapy for all HBsAg-positive patients with cirrhosis even if initially HBV DNA–negative to prevent future reactivation.

†Dosing should be according to renal status as indicated in drug brochure.

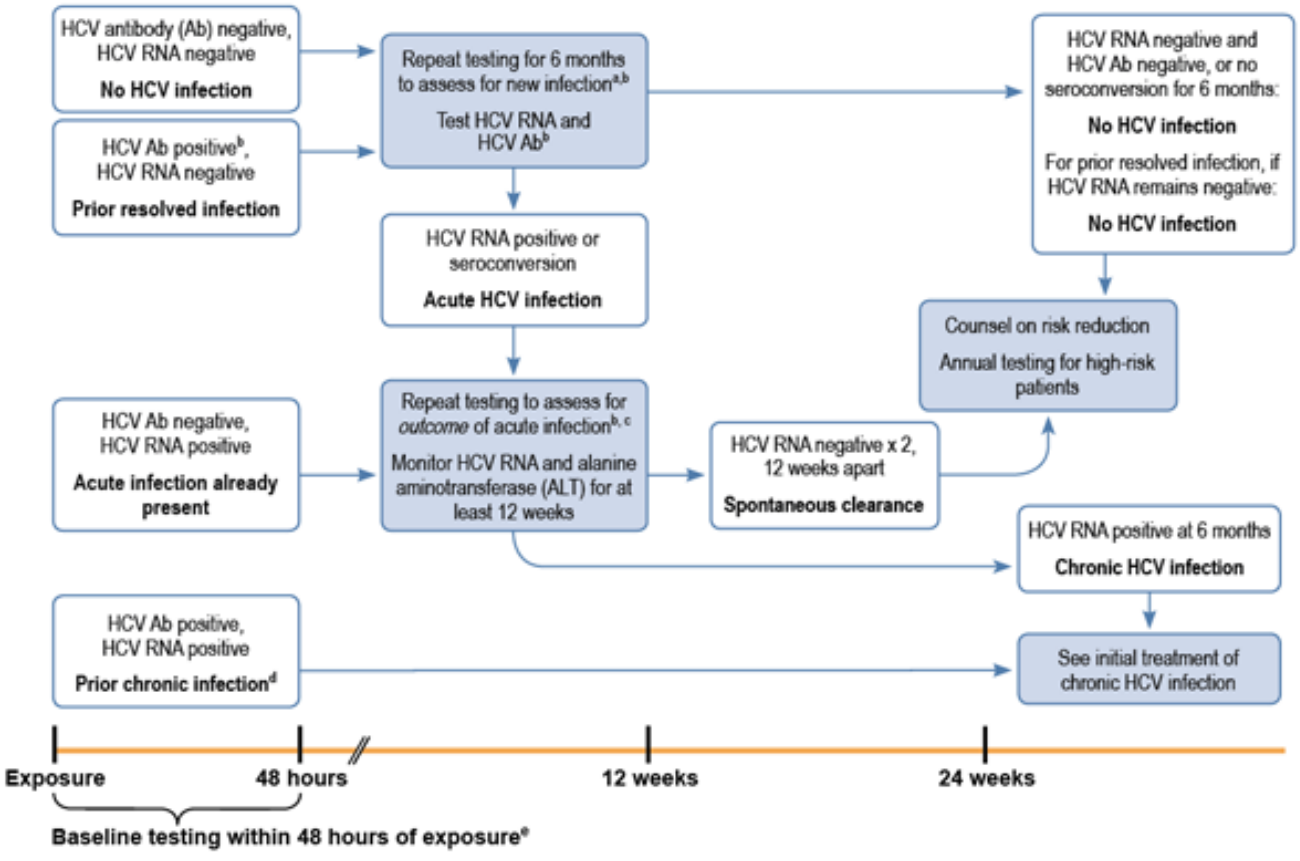
‡Rarely, such patients may present with <2,000 IU of HBV DNA following extensive hepatocytolysis. Maintenance antiviral therapy is strongly recommended, however, to prevent future reactivation episodes or ready the patient for transplantation. See the text for further details.

§Either agent can be used if the anticipated duration of therapy is ≤6 months.

Hepatitis C

APASL 2016 guideline

Figure. Testing Algorithm for Discrete Recognized Hepatitis C Virus (HCV) Exposure[a]



Treatment of Hepatitis C

Prevent HCV infection complication by achievement of sustained virologic response (SVR)

1. **Interferon**
2. **Ribavirin**
3. **direct-acting antivirals (DAAs)**

Alcoholic hepatitis

AminiM,etal,World J Gastroenterol 2010;16:4905–4912

Diagnosis: Based on history (Alcohol: 350~420g /week)and clinical and/or histological data

Prognosis score:

TABLE 1: Overview of the Formulas of the Prognostic Scores Discussed

Prognosis Score	Formula			
DF	$4.6 \times$ (patient PT − control PT) + bilirubin			
MELD	$3.8 \times \ln$(bilirubin (mg/dL)) + $1.2 \times \ln$(INR) + $9.6 \times \ln$(creatinine (mg/dL)) + 6.43			
GAHS		point	2 points	3 points
	Age	<50	≥50	---
	Leukocyte count (10^9/L)	<15	≥15	---
	Urea level (mmol/L)	<5	≥5	---
	PT ratio	<1.5	1.5-2.0	>2.0
	Bilirubin level (mmol/L)	<125	125-250	250
ABIC	(age $\times$ 0.1) + (serum bilirubin (mg/dL)) $\times$ 0.08) + (serum creatinine (mg/dL) $\times$ 0.3) + (INR $\times$ 0.8)			
Lille	exp(−R)/(1 + exp(−R))*			

*R = 3.19 − 0.101 × (age in years) + 0.147 × (albumin day 0 in g/L) + 0.0165 × (bilirubin day 0 − bilirubin day 7 (mmol/L)) − 0.206 × (renal insufficiency) − 0.0065 × (bilirubin day 0 in mmol/L) − 0.0096 × (PT in seconds).

Treatment of alcoholic hepatitis:

Prednisolone at 40 mg/day for 28 days for discriminant function >32 or with hepatic encephalopathy

Cochrane hepato-Biliary Group systematic review Aliment PharmacolTher 2008;27:1167–1178

Oralpentoxifylline (PTX) at 400 mg three times daily for 28 days is an alternative treatment for severe Alcoholic hepatitis by decreasing the incidence of hepatorenal syndrome

O' shea RS, Dasarathy S, McCullough AJ.Alcoholic liver disease.Hepatology 2010;51:307–328.

Autoimmune hepatitis

Diagnosis: based on a combination of clinical, biochemical, immunological, and histological features and the exclusion of other known causes of liver disease.Liver biopsy is needed to confirm the diagnosis

Manns MP, Czaja AJ, Gorham JD, Krawitt EL, Mieli-Vergani G, VerganiD,et al. Hepatology 2010;51:2193-2213

TABLE 2 Simplified Criteria for the Diagnosis of AIH

Variable	Cutoff	Points*
ANA or SMA	≥1:40	1
ANA or SMA	≥1:80	2†
or anti-LKM1	≥1:40	
or SLA	Positive	
IgG	>Upper limit of normal	1
	>1.10 times upper limit of normal	2
Liver histology	Compatible with AIH	1
	Typical of AIH	2
Absence of viral hepatitis	Yes	2

- score=6 indicates probable AIH; score>=7 indicates definite AIH

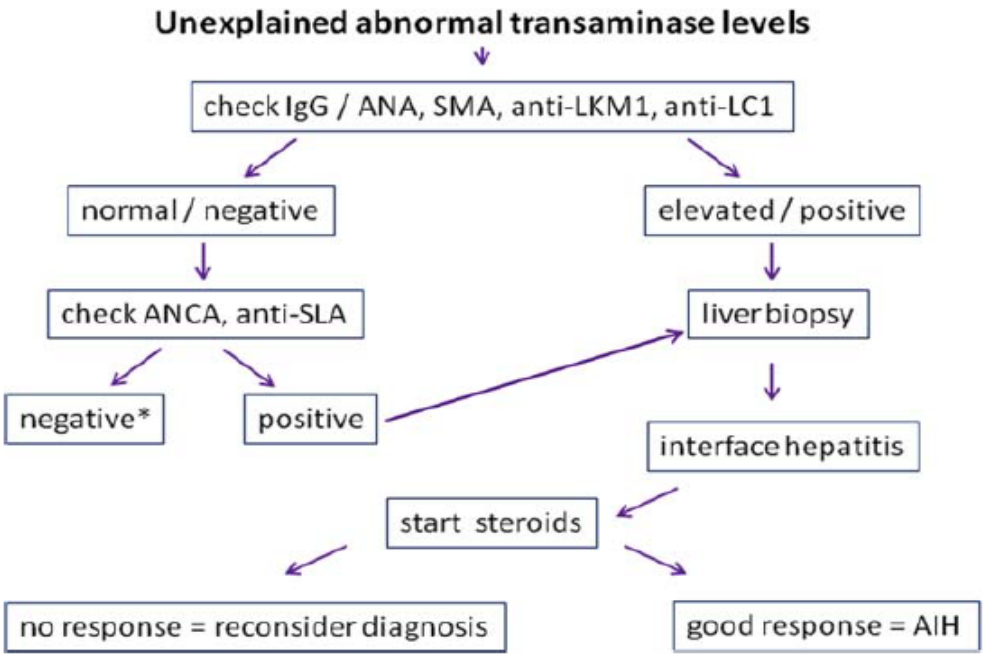


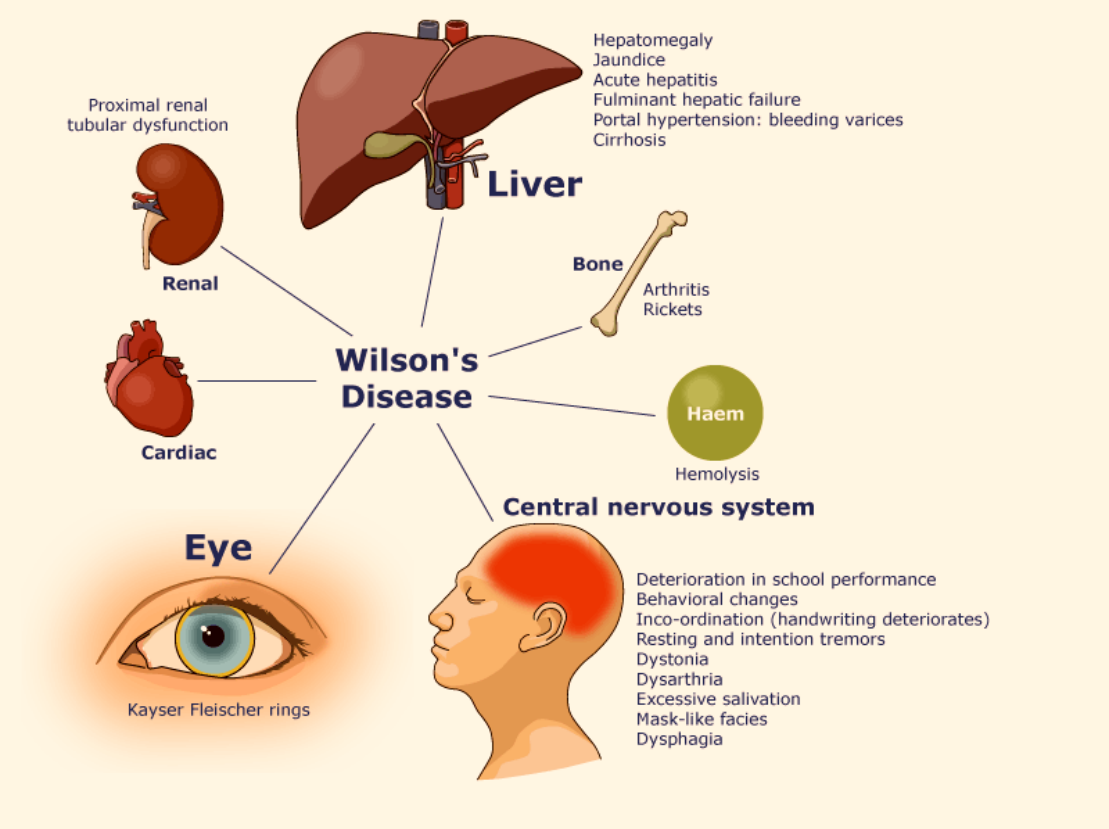
Figure 1 Diagnostic algorithm. *Observe for 2 to 3 months, but if serum aminotransferase levels remain elevated, consider liver biopsy and treat the patient if the biopsy sample shows interface hepatitis.

Drug induced liver injury

Diagnosis: hepatitisoccur within the first 6 months after drug initiation
Treatment: Discontinued medication,corticosteroids are not indicated unless a drug hypersensitivity such as the ‘drug rash with eosinophilia and systemic symptoms’ (DRESS) or autoimmune disease occur

Wilson's disease

Diagnosis:Age <40 years old. Obtain ceruloplasmin, serum and urinary copper levels, slit lamp examination for Kayser-Fleischer rings, hepatic copper levels when liver biopsy is feasible, and total bilirubin/ alkaline phosphatase ratio

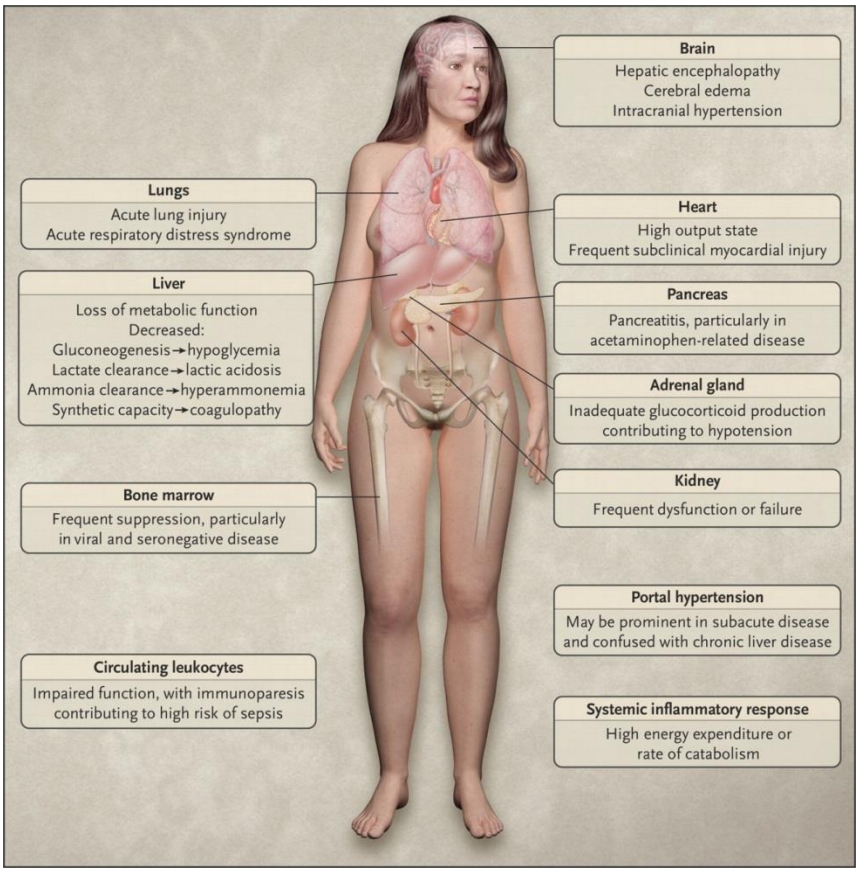


Acute liver failure

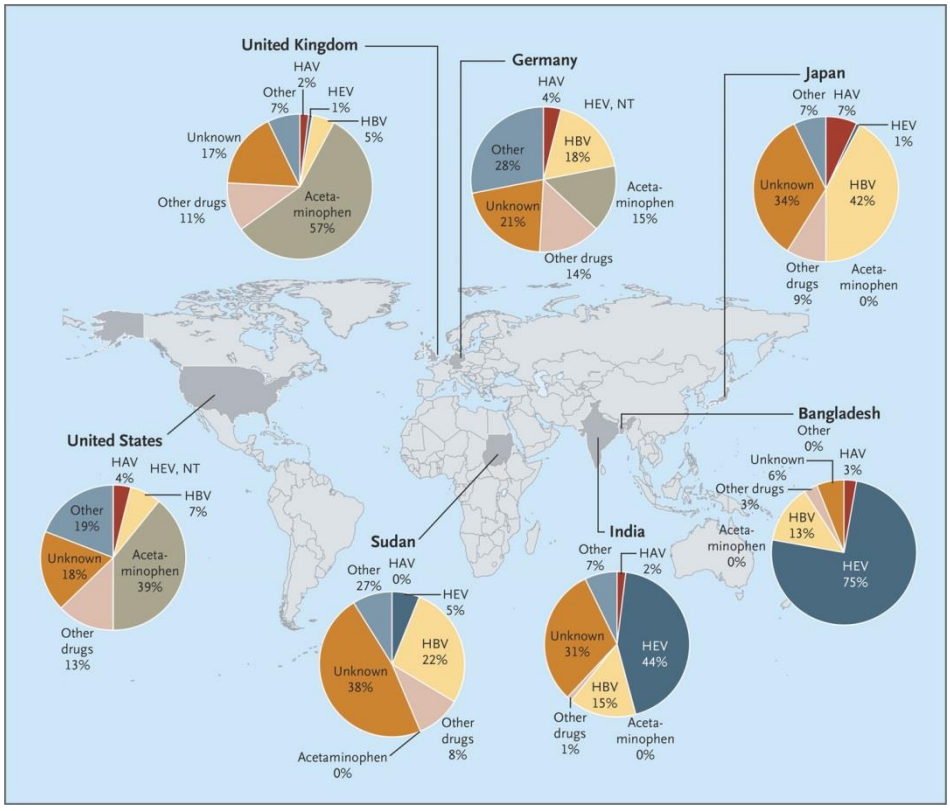
Definition: A patient without preexisting cirrhosis and with an illness of <26 weeks’ duration. includes both symptom:

- Evidence of coagulation abnormality, (INR) ≥1.5
- Any degree of mental alteration (encephalopathy)

N Engl J Med 2013;369:2525-2534



Etiology



Diagnosis and initial evaluation:

TABLE 2. INITIAL LABORATORY ANALYSIS

Prothrombin time/INR
Chemistries sodium, potassium, chloride, bicarbonate, calcium, magnesium, phosphate glucose AST, ALT, alkaline phosphatase, GGT, total bilirubin, albumin creatinine, blood urea nitrogen
Arterial blood gas
Arterial lactate
Complete blood count
Blood type and screen
Acetaminophen level
Toxicology screen
Viral hepatitis serologies anti-HAV IgM, HBsAg, anti-HBc IgM, anti-HEV[§], anti-HCV, HCV RNA*, HSV1 IgM, VZV
Ceruloplasmin level[#]
Pregnancy test (females)
Ammonia (arterial if possible)
Autoimmune Markers ANA, ASMA, Immunoglobulin levels
HIV-1, HIV-2[‡]
Amylase and lipase

**Done to recognize potential underlying infection.*
[#]Done only if Wilson disease is a consideration (e.g., in patients less than 40 years without another obvious explanation for ALF); in this case uric acid level and bilirubin to alkaline phosphatase ratio may be helpful as well.
[‡] Implications for potential liver transplantation.
[§] If clinically indicated.

Prognosis evaluation:

ETIOLOGY

Idiosyncratic drug injury
Acute hepatitis B (and other non-hepatitis A viral infections)
Autoimmune hepatitis
Mushroom poisoning
Wilson disease
Budd-Chiari syndrome

COMA GRADE ON ADMISSION

III or IV

KING'S COLLEGE CRITERIA

Acetaminophen-Induced ALF

- Strongly consider OLT listing if:
  - arterial lactate >3.5 mmol/L after early fluid resuscitation
- List for OLT if:
  - pH <7.3 – or –
  - arterial lactate >3.0 mmol/L after adequate fluid resuscitation
- List for OLT if all 3 occur within a 24-hour period:
  - presence of grade 3 or 4 hepatic encephalopathy
  - INR >6.5
  - Creatinine >3.4 mg/dL

Non-Acetaminophen-Induced ALF

- List for OLT if:
  - INR >6.5 and encephalopathy present (irrespective of grade)
  or any three of the following (encephalopathy present; irrespective of grade):
  - Age <10 or >40 years‡
  - Jaundice for >7 days before development of encephalopathy‡
  - INR ≥3.5
  - serum bilirubin ≥17 mg/dL
  - Unfavorable etiology, such as
    Wilson Disease, idiosyncratic drug reaction, seronegative hepatitis

Intensive care

guideline:

Cerebral Edema/Intracranial Hypertension
Grade I/II Encephalopathy Consider transfer to liver transplant facility and listing for transplantation Brain CT: rule out other causes of decreased mental status; little utility to identify cerebral edema Avoid stimulation; avoid sedation if possible Antibiotics: surveillance and treatment of infection required; prophylaxis possibly helpful Lactulose, possibly helpful
Grade III/IV Encephalopathy Continue management strategies listed above Intubate trachea (may require sedation) Elevate head of bed Consider placement of ICP monitoring device Immediate treatment of seizures required; prophylaxis of unclear value Mannitol: use for severe elevation of ICP or first clinical signs of herniation Hypertonic saline to raise serum sodium to 145-155 mmol/L Hyperventilation: effects short-lived; may use for impending herniation
Infection
Surveillance for and prompt antimicrobial treatment of infection required Antibiotic prophylaxis possibly helpful but not proven
Coagulopathy
Vitamin K: give at least one dose FFP: give only for invasive procedures or active bleeding Platelets: give only for invasive procedures or active bleeding Recombinant activated factor VII: possibly effective for invasive procedures Prophylaxis for stress ulceration: give H_2 blocker or PPI
Hemodynamics/Renal Failure
Volume replacement Pressor support (dopamine, epinephrine, norepinephrine) as needed to maintain adequate mean arterial pressure Avoid nephrotoxic agents Continuous modes of hemodialysis if needed Vasopressin recommended in hypotension refractory to volume resuscitation and no repinephrine
Metabolic Concerns
Follow closely: glucose, potassium, magnesium, phosphate Consider nutrition: enteral feedings if possible or total parenteral nutrition

Treatment:

- Metabolic and Nutritional Support
- N-Acetylcysteine
- Artificial support
  - molecular absorbents recirculation system (MARS)
  - Plasma exchange
- Bioartificial support
- G-CSF therapy
- Liver transplantation

Liver transplant evaluation for Acute Liver Failure:

TABLE 5 King's College Criteria

Acetaminophen Overdose	Nonacetaminophen Overdose
Arterial pH < 7.30 *or* all of the following: • Prothrombin time > 100 seconds (INR > 6.5) • Creatinine level > 3.4 mg/dL • Grade 3/4 encephalopathy	Prothrombin time > 100 seconds (INR > 6.5) *or* any three of the following: • Non-A, non-B hepatitis; idiosyncratic drug reaction; halothane etiology • Time from jaundice to encephalopathy > 7 days • Age < 10 years or > 40 years • Prothrombin time > 50 seconds (INR > 3.5) • Bilirubin level > 17.4 mg/dL

Hepatocellulcar carcinoma

- Diagnosis

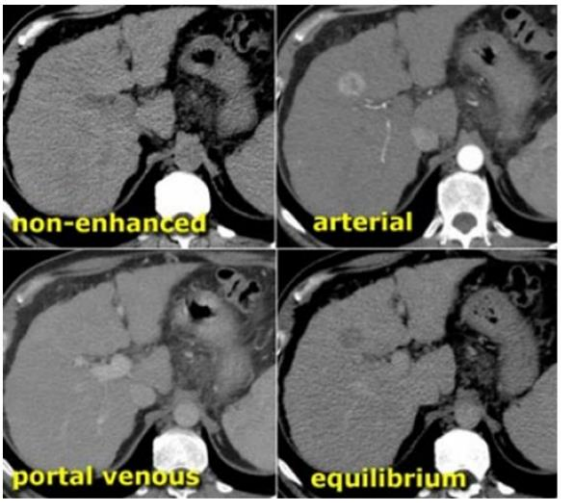
如果發現有≥1 cm 的 liver nodule 時（大部份是藉由腹部超音波發現），可以選擇 dynamic CT，dynamic MRI 或 EOB-MRI 來作診斷，如果在肝癌高風險族群（肝硬化，B 型肝炎或 C 型肝炎）有典型的 HCC 影像學特徵（arterial hyperenhancement, and wash-out in the venous and/or delayed phases, or hepatobiliary phase），就可以診斷為 HCC。如果上述的檢查無法排除惡性腫瘤且沒有典型的 HCC 影像學特徵時，則可改作另外一種檢查、contrast-enhanced ultrasound 或是直接 biopsy，如果第二次的影像檢查仍無定案時則可作 biopsy。

- AFP is no longer used for diagnosis of HCC because AFP can be elevated in some conditions without HCC (active hepatitis, cirrhosis, and intrahepatic cholangiocarcinoma and metastases from colon cancer. HCC can be diagnosed by imaging or/and tissue proof.

- 典型的 HCC 影像學特徵

前提：需在肝癌高風險族群才能作為診斷的依據（肝硬化，B 型肝炎或 C 型肝炎）

  - Non-enhanced phase: 當然是都不會亮。
  - Arterial phase: HCC brighter than the surrounding liver. HCC 相較於正常肝組織是以 artery 供應為主，而正常肝組織則是以 portal vein 供應為主，所以在 arterial phase，HCC 會比正常肝組織有較明顯的顯影。
  - Venous phase: 又稱為 portal venous phase，HCC less brighter than the surrounding liver. HCC 是以 artery 供應為主，而正常肝組織則是以 portal vein 供應為主，所以在 venous phase，HCC 會比正常肝組織有較不明顯的顯影。
  - Delayed phase: 也可稱為 wash-out or equilibrium phase，原理同 venous phase。
  - This phenomenon is known as “low-high-low-low” or “wash-in in the arterial phase and wash-out in the venous and/or delayed phases”.



- Staging
  - Barcelona Clinic Liver Cancer (BCLC) staging system

修訂後 BCLC 期別版本 (自108年起新診斷之肝癌個案適用)			
定義	Tumor Features	Child-Pugh Score	Performance Status Test
Stage 0	Single ≤ 2 cm Carcinoma in situ	Child-Pugh A	0-1
Stage A	**Single ≤ 5 cm*** or 3 nodulars ≤ 3 cm	Child-Pugh A-B	0-1
Stage B	**Single > 5 cm*** or Multinodulars	Child-Pugh A-B	0-1
Stage C	Portal invasion N1, M1	Child-Pugh A-B	0-2
Stage D	Any	Child-Pugh C	3-4

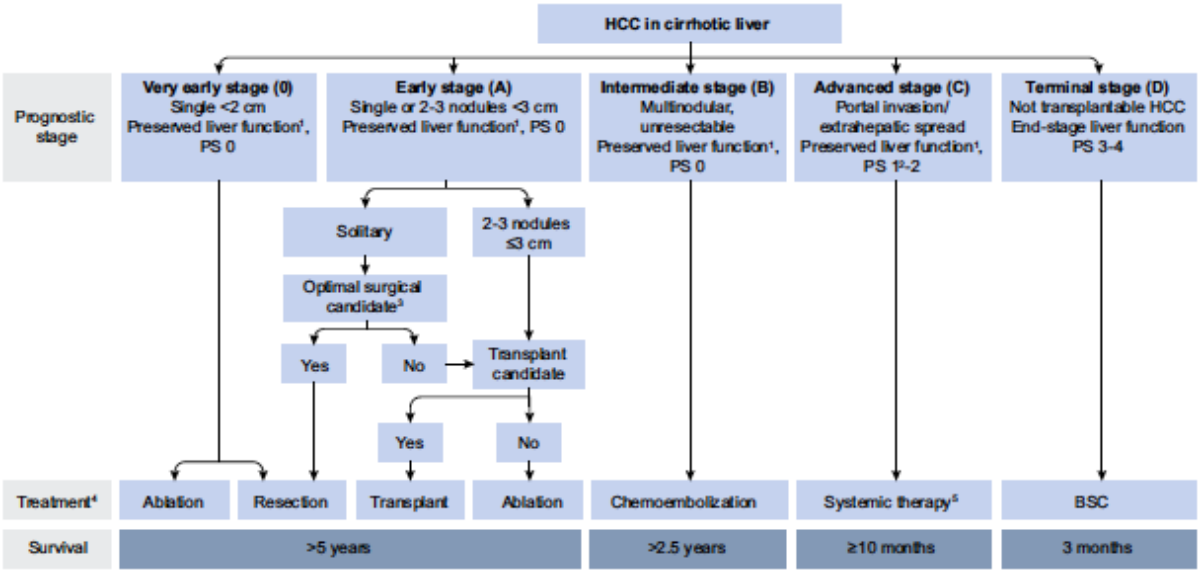
  - Tumor-node-metastasis (TNM) staging system (AJCC 8th)

	Primary tumor (T)	Regional lymph nodes (N)		Distant metastases (M)	
T1a	Solitary tumor ≤2 cm with/without vascular invasion	Nx	Regional lymph nodes cannot be assessed	M0	No distant metastasis
T1b	Solitary tumor >2 cm without vascular invasion	N0	No regional lymph node metastasis	M1	Distant metastasis
T2	Solitary tumor >2 cm with vascular invasion or multifocal tumors, none >5 cm	N1	Regional lymph node metastasis		
T3	Multifocal tumors at least one of which is >5 cm				
T4	Single tumor or multifocal tumors of any size involving a major branch of the portal vein or hepatic vein or tumor(s) with direct invasion of adjacent organs other than the gallbladder or with perforation of visceral peritoneum				
Stage					
Stage IA	T1a	N0	M0		
Stage IB	T1b	N0	M0		
Stage II	T2	N0	M0		
Stage IIIA	T3	N0	M0		
Stage IIIB	T4	N0	M0		
Stage IVA	Any T	N1	M0		
Stage IVB	Any T	Any N	M1		

- **Prognosis**
  - Median survival times of untreated patients
  - Very early stage (BCLC stage 0):
  - Early stage (BCLC stage A): ~36 months
  - Intermediate stage (BCLC stage B): ~16 months or 49% at 2 year
  - Advanced stage (BCLC stage C): 6-8 months or 25% at 1 year
  - Terminal stage (BCLC stage D): 3-4 months or 11% at 1 year

  - Median survival times of treated patients
  - Very early and early stages (BCLC stage 0 and 1): >5 years
  - Intermediate stage (BCLC stage B): >2.5 years
  - Advanced stage (BCLC stage C): ≥10 months
  - Terminal stage (BCLC stage D): 3-4 months or 11% at 1 year

- **Treatment**



- 基本上是按照上圖的 BCLC staging system 來選擇 treatment plan。另外也可按照下圖 2020 年台灣肝癌醫學會（Taiwan Liver Cancer Association, TLCA）更新的 consensus guideline 來選擇 treatment。

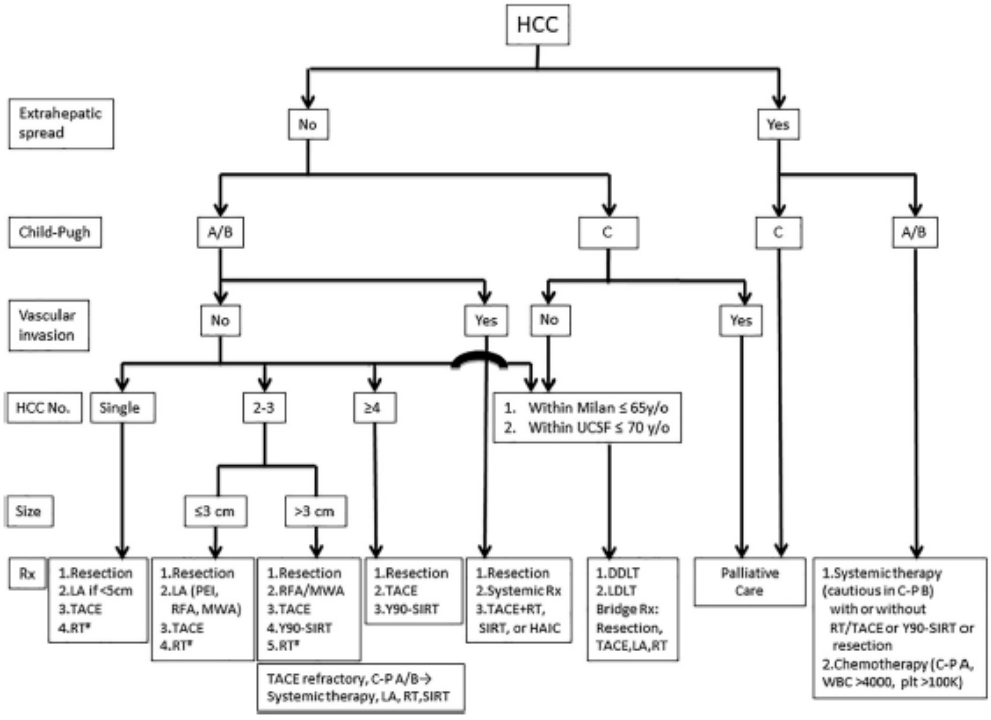


TABLE 111-5 TREATMENT OPTIONS FOR HEPATOCELLULAR CARCINOMA

Surgery
Resection
Liver transplantation
Local Ablative Therapies
Radiofrequency ablation (RFA)
Microwave ablation (MWA)
Cryosurgery
Percutaneous ethanol injection (PEI)
Regional Therapies: Hepatic Artery Transcatheter Treatments
Transarterial chemotherapy
Transarterial embolization
Transarterial chemoembolization
Transarterial drug-eluting beads
Transarterial radiotherapies:
^{90}Yttrium microspheres
^{131}Iodine– Ethiodol
Proton beam radiation
Conformal External-Beam Radiation and Intensity-Modulated Radiation Therapy
Systemic therapies
Molecularly targeted therapies (sorafenib, etc.)
Chemotherapy
Immunotherapy
Hormonal therapy + growth control
Supportive Therapies

- **Surgical resection**
  - Surgical resection is the treatment of choice for HCC in patients without cirrhosis or with compensated cirrhosis, or in selective patients complicated with cirrhosis and portal hypertension. Curative resection may also be performed in (A) multiple HCC located in one liver of liver, (B) HCC involving ipsilateral portal vein or hepatic vein, and (C) HCC with extrahepatic single organ involvement.
  - 5-year survival: >50%
  - Perioperative mortality: <5%
  - 5-year recurrence: 70%

- **Liver transplantation**
  - Patients with cirrhosis and a limited tumor burden (Milan criteria, and UCSF criteria in selected patients)
  - Milan criteria: A single tumor ≤5 cm or 2-3 tumors ≤3 cm
  - University of California San Francisco (UCSF) criteria: A single tumor ≤6.5 cm or 2-3 tumors ≤4.5 cm (total tumor diameter ≤8cm)
  - Within Milan criteria
  - 5-year and 10-year survival: 70% and 50%
  - 5-year recurrence: 10-15%

- **Radiofrequency ablation (RFA)**
  - RFA is the most used technique for local ablation, and has been advocated as a first-line curative therapy for very early stage (BCLC stage 0) or unresectable early stage (BCLC stage A) HCC. RFA uses a high-frequency alternating current generated from the tip of an electrode to produce coagulative necrosis of the tumor by the frictional heat.
  - Complete response: 70-90%
  - Median overall survival: ~60 months
  - 5-year recurrence: 50-70%

- **Transarterial chemoembolization (TACE)**
  - TACE is the most widely used treatment for unresectable HCC, and has been recommended as a first-line therapy for intermediate stage (BCLC stage B) HCC. The majority of the blood supply to HCC is derived from the hepatic artery, and the majority of inflow to the surrounding liver is through the poral vein. The rationale for TACE is that intra-arterial infusion of a cytotoxic agent (usually mixed with lipiodol) followed by embolization of the tumor-feeding vessels (usually with gel-foam particles) will result in a strong cytotoxic and ischemic effect targeted to HCC rather than the surrounding liver.
  - Median overall survical: ~30 months
  - Treatment-related mortality: 0.6%

- **Systemic therapies**
  - Systemic therapies 包含了標靶、免疫治療及化學治療藥物，主要使用於 advanced stage (BCLC stage C) 或不適合 locoregional therapies（surgical resection, RFA, and TACE）或效果不佳的 HCC。目前新藥的臨床試驗持續進行中，僅簡單介紹。第一線治療的藥物有 sorafenib, lenvatinib, and combination therapy (atezolizumab and bevacizumab)：sorafenib and lenvatinib 可用於 advanced stage (BCLC stage C)或 unresectable and not amenable to locoregional therapies，Combination therapy (atezolizumab and bevacizumab) 可用於 advanced stage (BCLC stage C) HCC。第二線治療的藥物有 regorafenib, ramucirumab (when AFP ≥400 ng/mL), and cabozantinib，目前證據僅限用於 sorafenib 使用後的病人，lenvatinib and combination therapy (atezolizumab and bevacizumab)目前尚無後續第二線的用藥。除此之外，nivolumab± ipilimumab, and pembrolizumab 可也嘗試作為第二線治療的藥物，目前證據僅限用於 sorafenib 使用後的病人，但證據力較上述三種 targeted therapies 薄弱。

Acute pancreatitis

Etiology

1. Most common: Gallstones (35%), significant history of alcohol use (30%), serum triglyceride >1,000mg/dl.
2. Others:
   a. A patient>40 years: a pancreatic tumor should be considered.
   b. Endoscopic investigation for acute idiopathic pancreatitis should be limited.
   c. Idiopathic pancreatitis => referred to centers of expertise
   d. Vascular etiology should be considered at vasculitis patient (SLE, polyarteritis nodosa)
   e. Genetic testing (SPINK1, PRSS1, CFTR) in young patients (<30 years old) if no cause is evident and a family history of pancreatic disease is present
   f. The other cause: miscellaneous(post-ERCP, pregnancy), infection, trauma, drug…etc.

TABLE 371-1 CAUSES OF ACUTE PANCREATITIS

Common Causes
Gallstones (including microlithiasis)
Alcohol (acute and chronic alcoholism)
Hypertriglyceridemia
Endoscopic retrograde cholangiopancreatography (ERCP), especially after biliary manometry
Drugs (azathioprine, 6-mercaptopurine, sulfonamides, estrogens, tetracycline, valproic acid, anti-HIV medications, 5-aminosalicylic acid [5-ASA])
Trauma (especially blunt abdominal trauma)
Postoperative (abdominal and nonabdominal operations)
Uncommon Causes
Vascular causes and vasculitis (ischemic-hypoperfusion states after cardiac surgery)
Connective tissue disorders and thrombotic thrombocytopenic purpura (TTP)
Cancer of the pancreas
Hypercalcemia
Periampullary diverticulum
Pancreas divisum
Hereditary pancreatitis
Cystic fibrosis
Renal failure
Infections (mumps, coxsackievirus, cytomegalovirus, echovirus, parasites)
Autoimmune (e.g., type 1 and type 2)
Causes to Consider in Patients with Recurrent Bouts of Acute Pancreatitis Without an Obvious Etiology
Occult disease of the biliary tree or pancreatic ducts, especially microlithiasis, biliary sludge
Drugs
Alcohol abuse
Metabolic: Hypertriglyceridemia, hypercalcemia
Anatomic: Pancreas divisum
Pancreatic cancer
Intraductal papillary mucinous neoplasm (IPMN)
Hereditary pancreatitis
Cystic fibrosis
Autoimmune
Idiopathic

Definition

TABLE 371-2 REVISED ATLANTA DEFINITIONS OF MORPHOLOGIC FEATURES OF ACUTE PANCREATITIS

Morphologic Feature	Definition	Computed Tomography Criteria
Interstitial pancreatitis	Acute inflammation of the pancreatic parenchyma and peripancreatic tissues, but without recognizable tissue necrosis	Pancreatic parenchyma enhancement by IV contrast agent No findings of peripancreatic necrosis
Necrotizing pancreatitis	Inflammation associated with pancreatic parenchymal necrosis and/or peripancreatic necrosis	Lack of pancreatic parenchymal enhancement by IV contrast agent and/or presence of findings of peripancreatic necrosis (see below—ANC and WON)
Acute pancreatic fluid collection	Peripancreatic fluid associated with interstitial edematous pancreatitis with no associated peripancreatic necrosis. This term applies only to areas of peripancreatic fluid seen within the first 4 weeks after onset of interstitial edematous pancreatitis and without the features of a pseudocyst.	Occurs in the setting of interstitial edematous pancreatitis Homogeneous collection with fluid density Confined by normal peripancreatic fascial planes No definable wall encapsulating the collection Adjacent to pancreas (no intrapancreatic extension)
Pancreatic pseudocyst	An encapsulated collection of fluid with a well-defined inflammatory wall usually outside the pancreas with minimal or no necrosis. This entity usually occurs >4 weeks after onset of interstitial edematous pancreatitis.	Well circumscribed, usually round or oval Homogeneous fluid density No nonliquid component Well-defined wall; that is, completely encapsulated Maturation usually requires >4 weeks after onset of acute pancreatitis; occurs after interstitial edematous pancreatitis

Acute necrotic collection (ANC)	A collection containing variable amounts of both fluid and necrosis associated with necrotizing pancreatitis; the necrosis can involve the pancreatic parenchyma and/or the peripancreatic tissues.	Occurs only in the setting of acute necrotizing pancreatitis Heterogeneous and nonliquid density of varying degrees in different locations (some appear homogeneous early in their course) No definable wall encapsulating the collection Location—intrapancreatic and/or extrapancreatic
Walled-off necrosis (WON)	A mature, encapsulated collection of pancreatic and/or peripancreatic necrosis that has developed a well-defined inflammatory wall. WON usually occurs >4 weeks after onset of necrotizing pancreatitis.	Heterogeneous with liquid and nonliquid density with varying degrees of loculations (some may appear homogeneous) Well-defined wall; that is, completely encapsulated Location—intrapancreatic and/or extrapancreatic Maturation usually requires 4 weeks after onset of acute necrotizing pancreatitis

Source: Modified from P Banks et al: Gut 62:102, 2013.

Diagnosis

1. 2 of the 3 following criteria: (i) abdominal pain consistent with the disease,(ii) serum amylase and/or lipase > 3x ULN, and/or (iii) characteristic findings from abdominal imaging
2. Other diagnostic tool:
   a.Abdominal ultrasound
   b. Contrast-enhanced computed tomographic (CECT) and/or magnetic resonanceimaging (MRI) if unclear diagnosis or who fail to improve clinically within the first 48-72h after admission

Initial assessment and risk stratification

1. Assess hemodynamic status and begin resuscitative measures.
2. Risk assessment to stratify higher- and lower-risk categories for further care.
3. Patients with organ failure => admission to intensive care unit

Table 3. Definitions of severity in acute pancreatitis: comparison of Atlanta and recent revision

Atlanta criteria (1993)	Atlanta Revision (2013)
Mild acute pancreatitis	**Mild acute pancreatitis**
Absence of organ failure	Absence of organ failure
Absence of local complications	Absence of local complications
Severe acute pancreatitis	Moderately severe acute pancreatitis
1. Local complications **AND/OR**	1. Local complications **AND/OR**
2. Organ failure	2. Transient organ failure (<48 h)
GI bleeding (>500 cc/24 hr)	**Severe acute pancreatitis**
Shock – SBP ≤90 mm Hg	Persistent organ failure >48 h[a]
PaO 2 ≤60 %	
Creatinine ≥2 mg/dl	

GI, gastrointestinal; SBP, systolic blood pressure.

[a]Persistent organ failure is now defined by a Modified Marshal Score (6,8)

Modified Marshall scoring system: definition of organ failure in revised Atlanta criteria (2013)

	Score				
Organ system	**0**	**1**	**2**	**3**	**4**
Respiratory (PaO_2/FiO_2)	>400	301–400	201–300	101–200	≤101
Renal*					
(serum creatinine, μmol/l)	≤134	134–169	170–310	311–439	>439
(serum creatinine, mg/dl)	<1.4	1.4–1.8	1.9–3.6	3.6–4.9	>4.9
Cardiovascular (systolic blood pressure, mm Hg)†	>90	<90, fluid responsive	<90, not fluid responsive	<90, pH<7.3	<90, pH<7.2

For non-ventilated patients, the FiO_2 can be estimated from below:

Supplemental oxygen (l/min)	**FiO_2 (%)**
Room air	21
2	25
4	30
6–8	40
9–10	50

A score of 2 or more in any system defines the presence of organ failure.

*A score for patients with pre-existing chronic renal failure depends on the extent of further deterioration of baseline renal function. No formal correction exists for a baseline serum creatinine ≥134 μmol/l or ≥1.4 mg/dl.

†Off inotropic support.

Initial management

1.Early aggressive hydration, defined as 250-500ml per hour of isotonic crystalloid solution the first 12-24h

- unless cardiovascular and/or renal comorbidites exist.

2. More rapid repletion (bolus) may be needed for severe volume depletion (hypotension and tachycardia)

3. Normal saline are bolused; Lactated Ringer’ s solution may be the preferred

4. Assess fluid requirements at frequent intervals within 6h of admission and for the next 24–48h.

5.The goal of aggressive hydration: decrease the blood urea nitrogen, U/O ≧ 0.5 mL/kg/h

- If a rise in BUN level, do a repeat volume challenge with a 2-L crystalloid bolus followed by increasing the fluid rate by 1.5 mg/kg per hour.

ERCP in acute pancreatitis

1. Acute pancreatitis with acute cholangitis => Do ERCP within 24h of admission.

2. ERCP is not needed in most patients with gallstone pancreatitis who lack laboratory or clinical evidence of ongoing biliary obstruction

3. Screen choledocholithiasis: MRCP or endoscopic ultrasound (EUS) rather than diagnostic ERCP

Table 1. Indications and Contraindications for ERCP in Patients with Acute Biliary Pancreatitis.

Indications
Suspected bile-duct stones as the cause of pancreatitis established clinically, and one of the following:
Cholangitis (fever, jaundice, sepsis)
Persistent biliary obstruction (conjugated bilirubin level >5 mg/dl [86 μmol per liter])
Clinical deterioration (worsening pain, increasing white-cell count, worsening vital signs)
Stone detected in the common bile duct on imaging
Contraindications
Absolute
Unstable medical condition precluding safe administration of moderate sedation or general anesthesia
Decision by competent patient not to provide consent for the procedure
Endoscopist with inadequate training in ERCP
Relative (may be overcome)
Anatomical condition (gastroduodenal disease or surgical alteration) that would impede endoscopic access to the major papilla; may be overcome in the case of a long Roux limb, for example, with the use of modified equipment and accessories
Clinically significant or uncorrectable coagulopathy; may be overcome, since a biliary stent can be placed without need for sphincterotomy

The role of antibiotics in acute pancreatitis

1. Antibiotics should be given for an extra-pancreatic infection, such as cholangitis, catheter-acquired infections, bacteremia, urinary tract infections, pneumonia

2. Not routine use of prophylactic antibiotics in patients with severe acute pancreatitis

3.Not use of antibiotics in patients with sterile necrosis to prevent the development of infected necrosis

4. Infected necrosis should be considered in patients with pancreatic or extra-pancreatic necrosis who deteriorate or fail to improve after 7–10 days
of hospitalization.

- In these patients, either (i) initial CT-guided fine needle aspiration (FNA) for Gram stain and culture to guide use of appropriateantibiotics or (ii) empiric use of antibiotics without CT FNA should be given.

5. Antibiotics known to penetrate pancreatic necrosis, such as carbapenems, quinolones, and metronidazole for infected necrosis

6. Not routine administration of antifungal agents along.

Nutrition in acute pancreatitis

1.Mild AP: oral feedings immediately if there is no nausea and vomiting, and abdominal pain has resolved

2. Mild AP: initiation of feeding with a low-fat solid diet or a clear liquid diet

3. Severe AP: enteral nutrition is recommended to prevent infectious complications.

- Not parenteral nutrition unless the enteral route isnot available, not tolerated, or not meeting caloric requirements

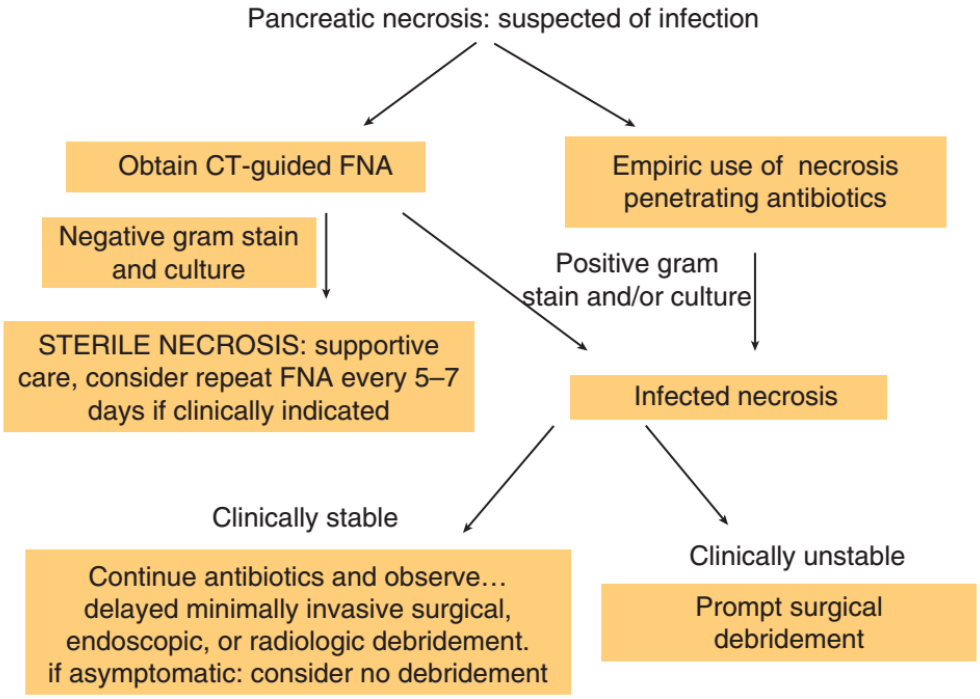
4. Pathway: oral, nasogastric delivery or naso-jejunal delivery of enteral feeding

The role of surgery in acute pancreatitis

1. Mild AP with gallstones→cholecystectomy before discharge.

2. Necrotizing biliary AP→cholecystectomy is to be deferred until active inflammation subsides and fluid collections resolve or stabilize

3. Asymptomatic pseudocysts and pancreatic and/or extra-pancreatic necrosis regardless of size, location and/or extension: no intervention
4. In stable patients with infected necrosis, surgical, radiologic, and/or endoscopic drainage should be delayed preferably for more than 4 weeks to allow liquification of the contents and the development of a fibrous wall around the necrosis (walled-off necrosis)
5. Symptomatic patients with infected necrosis: minimally invasive necrosectomy



Reference

1. American College of Gastroenterology Guideline:Management of Acute Pancreatitis, 2013
2.Japanese guidelines for the management of acute pancreatitis: JapaneseGuidelines 2015
3.Revised Japanese guidelines for the management of acute pancreatitis2015: revised concepts and updated points
4.Banks PA, Bollen TL, Dervenis C, Gooszen HG, Johnson CD, Sarr MG, et al. Acute Pancreatitis Classification Working Group. Classification of acute pancreatitis–2012: revision of the Atlanta classification and definitions by international consensus. Gut.2013
5. Evan L. Fogel, M.D., and Stuart Sherman, M.D. et al. ERCP for Gallstone Pancreatitis. N Engl J Med 2014; 370:150-157

Appendix: Severity score of acute pancreatitis

Ranson score

Criteria present at 0 hours:

Age >55 years old (1 point)

WBC >16,000/mm(3) (1 point)

Glucose >200 mg/dL (11.1 mmol/L) (1 point)

LDH >350 U/L (1 point)

AST >250 U/L (1 point)

Criteria present at 48 hours:

Hematocrit fall of 10% or greater (1 point)

BUN rise of 5 mg/dL or more (1.8 mmol/L) despite fluids (1 point)

Serum Calcium <8 mg/dL (2 mmol/L) (1 point)

pO2 <60 mmHg (1 point)

Base deficit >4 meq/L (1 point)

Fluid sequestration >6000 mL (1 point)

0-2 Points:	Mortality 0-3 percent
3-5 Points:	Mortality 11-15 percent
6-11 Points:	Mortality >=40 percent

Glagow-Imrie score (> 3 factors indicates severe pancreatitis)

Age > 55 years

White cell count > 15,000/mm^3

PaO2 < 60 mmHg

Serum lactate dehydrogenase > 600 U/L

Serum aspartate aminotransferase > 200 U/L

Serum albumin < 3.2 g/dL

Serum calcium < 8 mg/dL

Serum glucose > 180 mg/dL

Serum urea > 45 mg/dL

Bedside index of severity in acute pancreatitis (BISAP) score

BUN >25 mg/dL (8.9 mmol/L) (1 point)

Abnormal mental status with a Glasgow coma score <15 (1 point)

Evidence of SIRS (systemic inflammatory response syndrome) (1 point)

Patient age >60 years old (1 point)

Imaging study reveals pleural effusion (1 point)

0-2 Points:	Lower mortality (<2 percent)
3-5 Points:	Higher mortality (>15 percent)

SIRS score

Two or more of the following:

- Temperature >38° C or <36° C
- Heart rate >90 beats/min
- Respiratory rate >20 breaths/min or $PaCO_2$<32 torr
- WBC >12,000 cell/mm^3, <4,000 cells/mm^3, or >10% immature (band) forms

CT grading system (= CTSI, CT severity index):

Balthazar-Ranson grading system + pancreatic necrosis score

Grade	Balthazar Score	Points
A	Normal pancreas	0
B	Pancreatic enlargement	1
C	Pancreatic inflammation and/or peripancreatic fat	2
D	Single peripancreatic fluid collection	3
E	Two or more fluid collections and/or retroperitoneal air	4
	Percentage necrosis	
	0	0
	< 30	2
	30 – 50	4
	> 50	6
	CT Severity Index	
	Low degree	0 – 3
	Middle degree (6% mortality)	4 – 6
	High degree (17% mortality)	7 - 10

Biliary tract infection

Definition: acute inflammation and infection in the bile duct

Pathophysiology

(1) increased bacteria in the bile duct

(2) elevated intraductal pressure in the bile duct allowing translocation of bacteria

(3) endotoxin into the vascular and lymphatic system

Etiology:

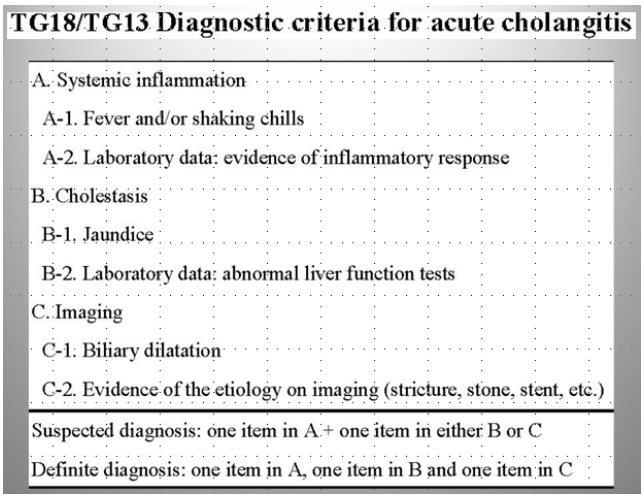
1. Cholelithiasis, 28~70%
2. Benign biliary stricture, 4~28%
3. Malignant biliary stricture, 10~57%
4. Primary sclerosing cholangitis, 0~24%
5. 罕見：Mirrizi syndrome (因腫大的膽囊壓迫到 CBD 使得膽道發炎),Lemmel syndrome(十二指腸憩室壓迫到 CBD 使得膽道發炎)

臨床症狀表現

- Charcot’ s triad: fever, RUQ pain, jaundice
- Reynold’ s pentad: fever, RUQ pain, jaundice, conscious change, shock

診斷

1. Charcot’ s triad: sensitivity 26.4%, specificity 95.9%
2. Tokyo Guideline 2013/2018 version: sensitivity 91.8%, specificity 77.7%
   - Definite diagnosis: one A + one B + one C
   - Suspected diagnosis: one A + one B or C



TG18/TG13 Diagnostic criteria for acute cholangitis

A. Systemic inflammation
A-1. Fever and/or shaking chills
A-2. Laboratory data: evidence of inflammatory response
B. Cholestasis
B-1. Jaundice
B-2. Laboratory data: abnormal liver function tests
C. Imaging
C-1. Biliary dilatation
C-2. Evidence of the etiology on imaging (stricture, stone, stent, etc.)
Suspected diagnosis: one item in A + one item in either B or C
Definite diagnosis: one item in A, one item in B and one item in C

嚴重度分級

- Grade III:Severe: any organ failure
- Grade II:Moderate: any two of following condition
- Grade I:Mild:Acute cholangitis does not meet the criteria of "Grade III (severe)" or "Grade II (moderate)"

TG18/TG13 Severity assessment criteria

Grade III (Severe) acute cholangitis

"Grade III" acute cholangitis is defined as acute cholangitis that is associated with the onset of dysfunction at least in any one of the following organs/systems

1.	Cardiovascular dysfunction	Hypotension requiring dopamine ≥5 μg/kg per min, or any dose of norepinephrine
2.	Neurological dysfunction	Disturbance of consciousness
3.	Respiratory dysfunction	PaO_2/FiO_2 ratio <300
4.	Renal dysfunction	Oliguria, serum creatinine >2.0 mg/dL
5.	Hepatic dysfunction	PT-INR >1.5
6.	Hematological dysfunction	Platelet count <1,00,000/mm^3

Grade II (moderate) acute cholangitis

"Grade II" acute cholangitis is associated with any two of the following conditions:

1. Abnormal WBC count (>12,000/mm^3, <4,000/mm^3)
2. High fever (≥39 °C)
3. Age (≥75 years)
4. Hyperbilirubinemia (total bilirubin ≥5 mg/dL)
5. Hypoalbuminemia (<STD × 0.7)

Grade I (mild) acute cholangitis

"Grade I" acute cholangitis does not meet the criteria of "Grade III (severe)" or "Grade II (moderate)" acute cholangitis at initial diagnosis

常見膽管炎菌種

- GNB: E.coli, K.p, pseudomonas, enterobacter…
- GPC: enterococcus, streptococcus, staphylococcus
- Anaerobes: 少見

治療

- 抗生素使用：不需要經驗使用"抗厭氧菌"抗生素，除非有 biliary enteric anastomosis 存在

1. Mild: Unasyn + aminoglycoside, cefazolin, ertapenem, ciprofloxacin or levofloxacin
2. Moderate: Tazocin, ceftriaxone, ertapenem, iprofloxacin or levofloxacin
3. Severe: Tazocin, cefepime or fortum, imipenem

- 不管嚴重度都需要做膽道的引流

1. mild:可先藥物治療，治療 24 小時後，反應不好的話還是要做引流
2. moderate: early biliary drainage, 24~48hr，要密切追蹤是否變成 severe
3. Severe: urgent biliary drainage, 24hr 內

引流方法

1. 內視鏡引流是首選方法
2. 皮下引流是替代方法
3. 開刀引流是最後手段，適用於“無法內視鏡或皮下引流的病人”或是“引流效果差的病人”

Management of Non-variceal UGI bleeding

【Etiology】

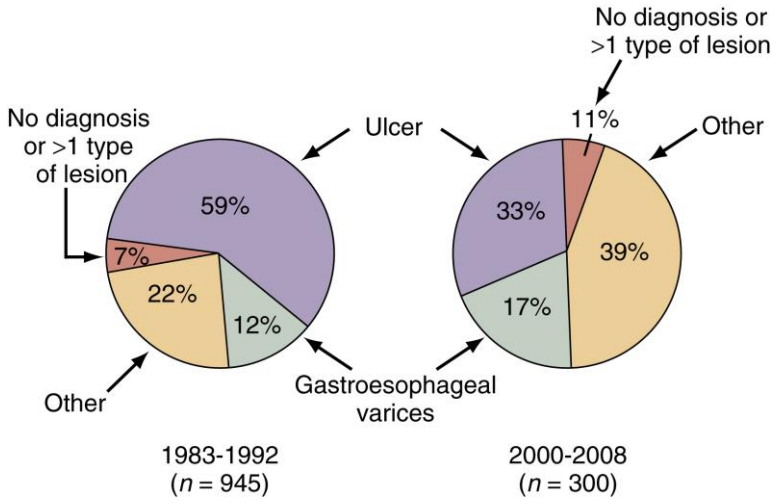
TABLE 20.2
Causes of Severe UGI Bleeding in the UCLA CURE Database (N *n* = 968)

Cause	Frequency (%)
Peptic ulcer	35.2
Esophageal or gastric varix	21.9
Portal hypertension-related lesion *	4.6
Esophagitis	4.6
Angioectasia †	4.0
Mallory-Weiss tear	4.0
Dieulafoy lesion	3.2
UGI tract neoplasm	3.1
Epistaxis	2.2
Erosions	1.2
Other	8.8
No cause found	7.3

CURE , Center for Ulcer Research and Education; *UCLA* , University of California, Los Angeles.

* Other than an esophageal or gastric varix

Risk factors of peptic ulcer disease



Sleisenger and Fordtran's table

【Resuscitation】

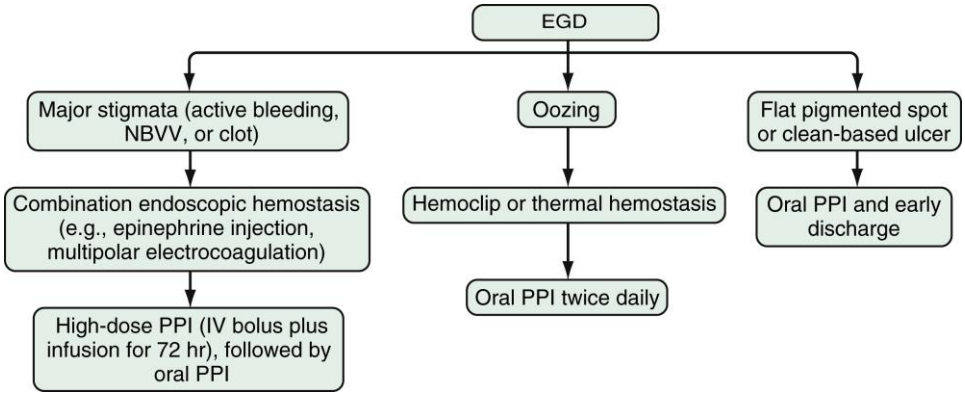
1. Maintain vital signs stable
2. Maintain Hb >7, PLT > 50000, PT (INR) < 2 or PT < 15 sec

	Class I	Class II	Class III	Class IV
Volume Loss (ml)	0–750	750–1500	1500–2000	>2000
Loss (%)	0–15	15–30	30–40	>40
RR	14–20	20–30	30–40	>40
HR	<100	>100	>120	>140
BP	Unchanged	Unchanged	Reduced	Reduced
Urine Output (ml/hr)	>30	20–30	5–15	Anuric
Mental State	Restless	Anxious	Anxious/confused	Confused/lethargic

3. Nasogastric tube insertion and gastric lavage
   (Evaluation the severity of bleeding)
4. Special considerations:

Patients with uremia or coronary artery diseases, to raise hematocrit at least >30% shall be beneficial

【Algorithm for endoscopic and medical management of severe peptic ulcer hemorrhage following hemodynamic stability】



【Pharmacological therapy】

1. High dose Proton pump inhibitor
   (EX: Esomeprazole 80mg bolus then, 8mg/hr IVF)
2. Hold anticoagulation

【UGI endoscopy】

1. Prepare: gastric lavage before UGI endoscopy
2. May use Erythromycin or Metoclopramide for prepare
   a. Erythromycin: 250 mg IV given 30-120 minutes prior to endoscopy

b. Metoclopramide: 1 amp IV given 30-120 minutes prior to endoscopy

3. Endoscopy evaluation score

Endoscopic Stigma (Forrest Class)	Frequency (%)	Risk of Rebleeding (%)	Risk of Rebleeding After Endoscopic Hemostasis (%) *
Active arterial bleeding (IA)	12	90	15-30
Nonbleeding visible vessel (IIA)	22	50	15-30
Adherent clot (IIB)	10	33	0-5
Oozing without stigmata (IB)	14	10-14 †	0-5
Flat spot (IIC)	10	10-25 †	0
Clean base (III)	32	3	N/A

4. Endoscopic management:

A. Forrest Ia, Ib, IIa:

a. Combination therapy
(epinephrine injection + contact thermal/mechanical)

b. High dose PPI after treatment for 72 hours

c. Survey and eradication Helicobacter pylori

d. May start clear liquid diet after endoscopy

B. Forrest IIb:

a. Clot remove then combine therapy/high dose PPI

b. High dose PPI after treatment for at least 3 days

c. Survey and eradication

d. May start clear liquid diet after endoscopy

C. Forrest IIc and III

a. No endoscopic hemostasis required

b. Once daily oral PPI

c. Start regular diet

d. Survey and eradication Helicobacter pylori

【Endoscopic hemostasis】

1. Injection therapy:

- Local tamponade from a volume effect
- Diluted epinephrine (1: 10000 or 1: 20000) injected 0.5~2 ml in and around the ulcer

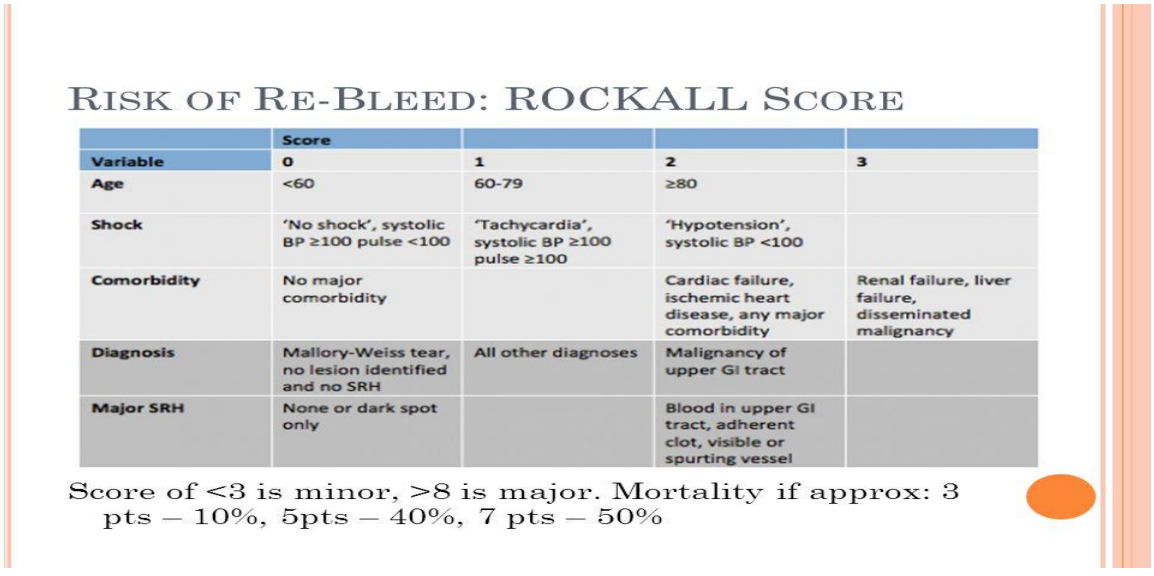
2. Thermal therapy:
   - Contact
     multipolar/bipolar electrocautery probe
   - Non-contact -> Argon plasma coagulation
3. Mechanical therapy:
   - Clips
   - Band ligation devices

【Post endoscopy care】

<<Re-bleeding risk assessment>>

1. High- risk populations of peptic ulcer bleeding and recurrent bleeding, including liver cirrhosis, end stage renal disease, probably chronic obstructive pulmonary disease, and type II diabetes.
2. NSAID and high-affinity serotonin reuptake inhibitors increase the risk of peptic ulcer bleeding.

3. Rockall scoring system combining both clinical and endoscopic variables may identify patients who are given early discharge or outpatient management who are in need of more aggressive treatment interventions, including endoscopy, and who have further bleeding or death



Gut. 2018 Mar; 67(3): 405–417.

Single antiplatelet agent

1. Withholding aspirin before endoscopy in aspirin users presenting with serious or life-threatening GI bleeding.

2. Early resumption of aspirin, preferably within 3–5 days after endoscopic haemostasis.

Dual antiplatelet therapy (DAPT)

1. Not Withholding Both antiplatelet agents due to high risk of stent thrombosis.
2. For patients on proton-pump inhibitor infusion and DAPT with aspirin and clopidogrel ➔ continuing aspirin and withholding clopidogrel.
3. In patients with drug-eluting coronary stents ➔ early resumption of P2Y12 receptor inhibitor, preferably within 5 days after endoscopic haemostasis.

Warfarin

1. Withholding warfarin to facilitate haemostasis.
2. 4-factor prothrombin complex concentrate (PCC) plus low-dose vitamin K for life-threatening bleeding with an INR above 2.5.
3. Not delaying endoscopy for life-threatening bleeding until normalisation of INR.
4. In patients with high thromboembolic risk ➔ not recommend higher doses of vitamin K (>5 mg).
5. In patients with high thromboembolic risk ➔ resume warfarin once adequate haemostasis is achieved.
6. In patients with low thromboembolic risk➔ bridging anticoagulation therapy is Not recommend
7. In patients with high thromboembolic risk➔ bridging anticoagulation with unfractionated heparin

Direct oral anticoagulants (DOACs)

1. Withholding DOACs to facilitate haemostasis.
2. Activated charcoal for life-threatening bleeding if the last dose of DOAC is taken within 3 hours.
3. Idarucizumab for the treatment of life-threatening bleeding in patients on dabigatran.

4. Resuming DOACs after adequate haemostasis is achieved.

5. In patients on DOACs ➔ bridging therapy is Not recommend

<<NSAID use>>

1. Avoid if possible
2. If necessary, use one of the following:

   (Decrease bleeding rate)

   a. COX-2 inhibitor + PPI

   b. NSAID + PPI

   c. COX-2 inhibitor

Treatment of Helicobacter pylori

- 1982 年，澳大利亞科學家，巴里·馬歇爾（Barry J. Marshall）和羅賓·沃倫（J. Robin Warren）自胃液中成功培養分離出幽門螺旋桿菌(Helicobacter pylori)，並進一步證實人體的胃潰瘍、胃炎等疾病是和該種細菌在胃部繁殖有關。打破了長久以來醫學界所認為的人體胃部強酸下不會有細菌長期生存；胃潰瘍、胃炎等疾病產生是單純由刺激性食物、緊張壓力等造成的觀念。
- 目前公認幽門螺旋桿菌感染是慢性活動性胃炎、消化性潰瘍、胃黏膜相關淋巴組織淋巴瘤和胃癌的主要致病因素。世界衛生組織更在 1994 年把幽門螺旋桿菌列為第一級致癌物。
- H. pylori has spiral shape and multiple unipolar flagella allow them to move freely through the gastric mucous layer. It would produce large amounts of urease, an enzyme that hydrolyzes urea to alkaline ammonia and CO 2 , permitting the bacteria to modulate the local pH.
- Indication for H. pylori survey and treatment:
    1. Uninvestigated Dyspepsia
    2. Gastroesophageal reflux disease with long-term PPI treatment
    3. NSAID/Aspirin user
    4. Unexplained iron deficiency anemia
    5. Idiopathic thrombocytopenic purpura
    6. Vitamin B12 deficiency
    7. Prevention for gastric cancer
    8. MALT lymphoma
    9. Peptic ulcer disease (gastric or duodenal ulcer)
- Diagnosis methods
- 臨床上診斷此菌的方法可以分為侵襲性及非侵襲性的方法。侵襲性的方法必須於胃鏡檢查時進行，包括病理組織學切片檢查、快速尿素酶檢查、細菌培養等。非侵襲性的方法為：碳十三或十四尿素呼吸法(UBT)、血清抗體檢查、糞便抗原檢查。
- 尿素呼氣法是目前確認根除與否最好的方法，糞便抗原檢查是另一個選擇但準確度較尿素呼氣法稍低。而血清抗體檢查不能區分出病人是否接受過治療，因為即使已治療成功，細菌不存在，血清抗體仍然可能呈現陽性。細菌培養主要是在幽門螺旋桿菌根除治療失敗後才實施，用以了解抗生素之抗藥性。

	No added Pathology cost	in patients using PPIs
Culture	Specificity ≈ 100% Allows antibiotic sensitivity testing	Difficult culture protocol Not widely available Expensive
PCR assay	Excellent sensitivity and specificity Permits detection of antibiotic resistance	Not widely available Technique not standardized Expensive

- Recent use of PPIs (within 2 weeks) or antibiotics (within 4 weeks) may lead to a decrease in the gastric bacterial load causing false-negative results of UBT and stool antigen test

第一線治療

一、 標準三合療法

標準三合療法：質子幫浦阻斷劑一顆，一日一次至二次 + clarithromycin 500 毫克一日二次 + amoxicillin 1 公克 (或 metronidazole 500 毫克)，一日二次。除菌的療程為 7 至 14 天。根據文獻的分析，14 天之三合療法比 7 天之三合療法約可提高 5%-12%的除菌率。

二、 系列性治療（Sequential Therapy）

系列性治療是先給病人 5 天的質子幫浦阻斷劑和 amoxicillin 的二合治療(質子幫浦阻斷劑一顆，一日二次+amoxicillin 1 公克，一日二次)，而後再給 5 天質子幫浦阻斷劑、clarithromycin 和 metronidazole 之三合治療(質子幫浦阻斷劑一顆，一日二次 + clarithromycin 500 毫克 + amoxicillin 1 公克，一日二次)

三、不含鉍劑之四合療法

質子幫浦阻斷劑一顆，一日二次 + clarithromycin 500 毫克，一日二次 + amoxicillin 1 公克，一日二次 + metronidazole 500 毫克，一日二次。除菌的療程為 7 至 10 天。

四、標準四合療法

質子幫浦阻斷劑一顆，一日二次；鉍劑 (如 bismuth subcitrate 120 毫克)，一日四次；metronidazole 500 毫克，一日三次；tetracycline 500 毫克，一日四次，療程為 7 至 14 天。

第二線治療

以 Levofloxacin 為基礎之三合療法: 質子幫浦阻斷劑一顆，一日二次 + levofloxacin 500 毫克，一日一次 + amoxicillin 1 公克，一日二次，療程為 7 至 14 天。然而 levofloxacin 抗藥性的快速增加是臨床上一個棘手的問題。

第三線治療

依「第四屆馬茲垂克治療共識」的建議，在作幽門螺旋桿菌的第三線治療時，宜依據幽門螺旋桿菌的抗生素敏感性試驗結果選擇抗生素。

- 由於幽門螺旋桿菌對於 clarithromycin 的抗藥性，是第一線標準三合療法失敗的主因，因而馬茲垂克治療共識中建議，需依當地的抗藥性狀況，選

擇第一線的治療方式。如果當地對於 clarithromycin 的抗藥性低於 15~20%，可選擇標準三合療法作為第一線治療。而在第二線治療時，可選擇以 Levofloxacin 為基礎之三合療法或含鉍劑之標準四合療法。如果當地對於 clarithromycin 的抗藥性已經高於 15~20%，因為標準三合療法的根除率已經不佳，需改以選擇系列性治療、標準四合療法或不含鉍劑之四合療法作為第一線治療。而在第二線治療時，可選擇以 Levofloxacin 為基礎之三合療法

- Treatment regimens (Sleisenger):

Regimen	Duration	Eradication Rate	Comments
Triple therapy: PPI, clarithromycin, and amoxicillin (each twice daily)	10-14 days	70%-85%	Macrolide resistance affects the eradication rate, and this regimen is not appropriate for regions with clarithromycin resistance>15%-20%
Triple therapy (with metronidazole): PPI, clarithromycin, and metronidazole (each twice daily)	10-14 days	70%-85%	Appropriate for penicillin-allergic individuals
Sequential therapy: PPI and amoxicillin for 5-7 days followed by PPI, clarithromycin, and nitroimidazole for 5-7 days (each twice daily)	10-14 days	75%-95%	Appears to be effective despite clarithromycin resistance; limited experience to date in the USA Either metronidazole or tinidazole may be used as the nitroimidazole
Quadruple therapy: PPI (twice daily) plus bismuth subsalicylate, metronidazole, and tetracycline (each 4 times daily)	10-14 days	75%-90%	Inexpensive but complex

- High dose PPI (twice a day) is more effective than standard dose for eradicating HP infection. “In vitro” studies show that antibiotic minimum inhibitory concentration is affected by gastric pH

<First-line treatment>:

- 14-day clarithromycin-containing triple therapy was more effective than

10- and 7-day regimens yielding an overall eradication rate >80%

- The standard 10-day sequential therapy has shown high efficacy in first-line HP treatment yielding eradication rates of about 90%
- If available, an alternative first-line treatment maybe the '3-drug pill (i.e., bismuth, metronidazole and tetracycline).A large multicenter European RCT with this regimen reported eradication rates >90%, even in patients with clarithromycin-resistant strains

<Second-line treatment>:

- Either bismuth-containing quadruple therapy or 10-day levofloxacin-containing triple therapy
- Triple therapy with PPI + levofloxacin + amoxicillin was not inferior in terms of efficacy to the more complex bismuth-containing quadruple therapy, providing cure rates of 88%

<Third-line treatment>:

- HP culture to allow a better choice of antibacterial treatment based on the antimicrobial resistance pattern
- 90–95% HP eradication rate using levofloxacin-amoxicillin triple therapy and bismuth-containing quadruple therapy as second- and third-line regimens. Therefore, after failure of second-line treatment with 10-day levofloxacin triple therapy, bismuth-containing quadruple therapy should be used as third-line treatment whenever bismuth salts are available
- A rifabutin-based regimen should be used in the treatment of refractory HP infection, namely in patients in whom all previous treatments failed (either 150 mg twice a day or 300 mg once a day for 10 days)

Inflammatory bowel disease

前言：

Inflammatory bowel disease (IBD)分為 Crohn’s disease (CD)和 Ulcerative colitis (UC)。以下先介紹兩者的基本差異，接下來分開介紹病人住院時應如何做初步的診斷、檢查與治療。

什麼是發炎性腸道疾病(IBD)?

發炎性腸道疾病(IBD)是一種自體免疫造成慢性腸道發炎的疾病，可分為潰瘍性結腸炎(UC)及克隆氏症(CD)。潰瘍性結腸炎只侵犯大腸，表淺黏膜層，造成連續性發炎病灶，大腸全切除可治癒。然而，克隆氏症則侵犯整個消化道 (口腔到肛門)，造成全層，跳躍性發炎病灶，導致嚴重併發症，比起潰瘍性結腸炎，也更不容易控制。

病因為何?

發炎性腸道疾病是由於基因、環境、飲食等因素，導致免疫系統失調，白血球攻擊自身正常的腸胃道組織，造成發炎、潰瘍甚至是出血。這個疾病過去好發於西方人，但是亞洲國家在過去十年來發生率大幅上升，隨著現代化的腳步，從日本、韓國到台灣個案數逐年增加。然而，我們的基因並沒有改變，因此最可能的病因還是跟飲食西化或食品添加物有關。

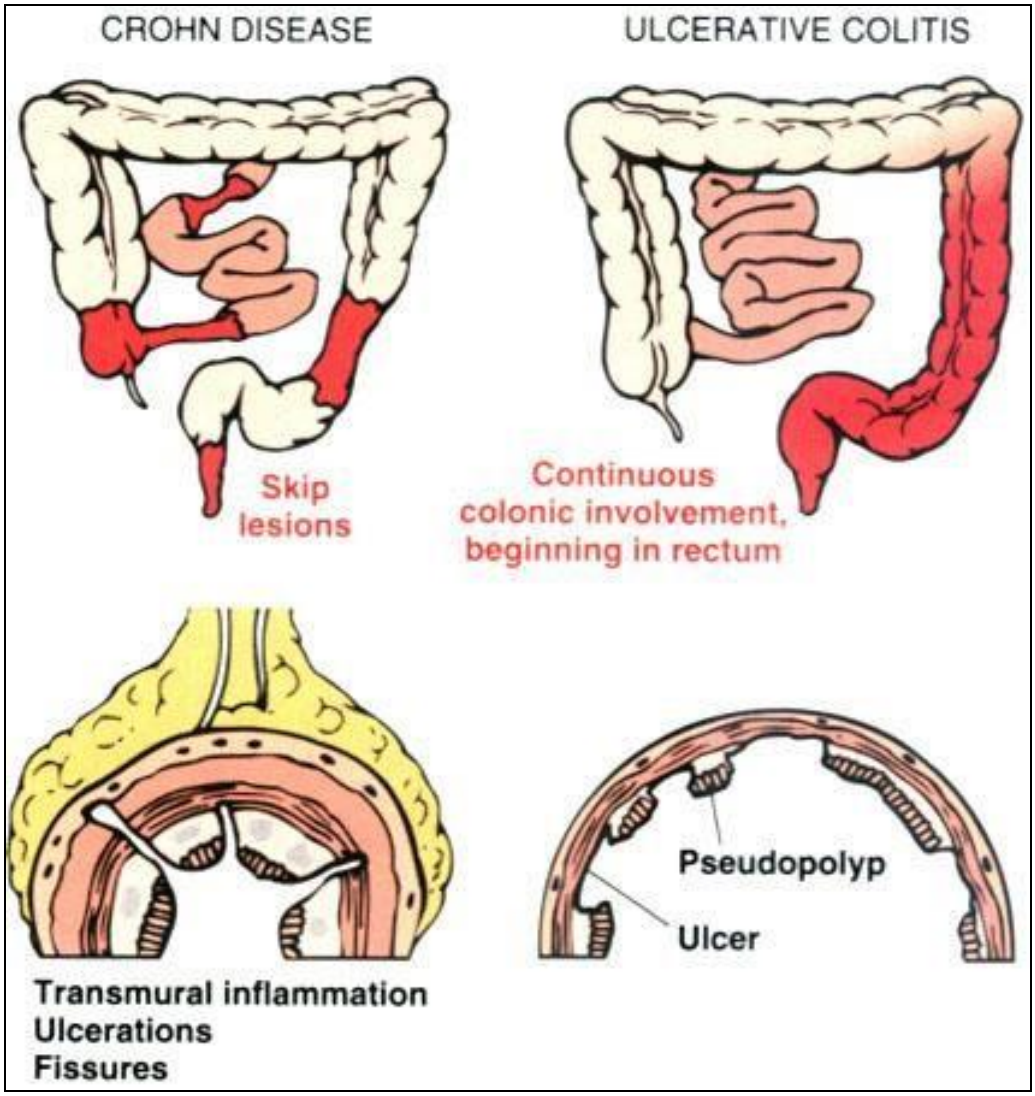
好發族群及臨床表現?

發炎性腸道疾病好發於 20 至 40 歲之間的年輕族群，其中一小部份發生在 60-80 歲的長者。常見腸胃道症狀包含拉肚子大於四周、肚子痛、便血、排便後仍有大不乾淨的感覺、貧血以及體重減輕等。克隆氏症較常侵犯迴盲瓣附近，容易造成穿孔、廔管、腹腔膿瘍、狹窄以及腹痛; 潰瘍性結腸炎最常侵犯直腸，容易造成裡急後重以及出血等症狀，發生大腸癌的風險也比克隆氏症高。則此外，部分患者有腸道以外的症狀(EIM)，包含肌肉骨骼(peripheral arthralgia、axial arthropathy、osteopenia)、皮膚黏膜(Pyoderma gangrenosum、erythema nodosum)、眼睛(uveitis)、膽胰(Primary sclerosing cholangitis)、腎臟、生殖器、血管等系統侵犯。更有些人會合併其他自體免疫疾病，包含僵直性脊椎炎、類風溼性關節炎和紅斑性狼瘡等等。

EIM	Parallel Course of IBD	Separate Course of IBD	May or May Not Parallel Disease Activity
Axial arthropathy		✓	
Peripheral arthropathy	✓ (Type I)	✓ (Type II)	
Erythema nodosum	✓		
Pyoderma gangrenosum			✓
Sweet's syndrome	✓		
Oral aphthous ulcers	✓		
Episcleritis	✓		
Uveitis			✓
PSC			✓

Adapted from Trikudanathan et al.[2]

- **CD 與 UC 常見區分方式，並非絕對，診斷仍需綜合判斷**



Feature	Crohn’ s Colitis	Ulcerative Colitis
Mucosal lesions	Aphthous ulcers are common in early disease; late disease is notable for stellate, rake, bear-claw, linear, or serpiginous ulcers and cobblestone like appearance	Micro-ulcers are more common, but larger ulcers are seen Pseudopolyps are more common
Distribution	Often discontinuous and asymmetric, with skipped segments of normal intervening mucosa, especially in early disease	Continuous, symmetric, and diffuse, with granularity or ulceration found in the entirety of involved segments; however, peri-appendiceal inflammation (cecal patch) is common, even when the cecum is not involved
Rectum	Complete, or more often relative, rectal sparing	Typically involved with variable proximal distribution
Ileum	Often involved (≈75% of	Not involved, except as

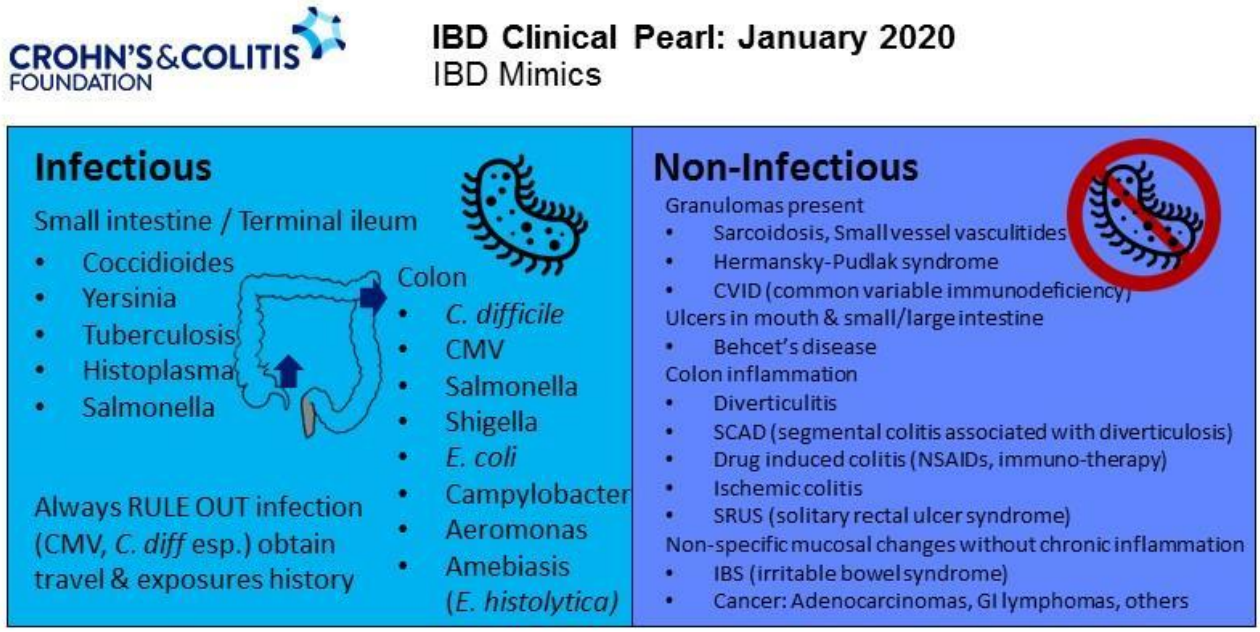
Feature	Crohn’ s Colitis	Ulcerative Colitis
	cases)	backwash ileitis in panulcerative colitis
Depth of inflammation	Mucosal, submucosal, and transmural	Mucosal; transmural only in fulminant disease
Serosal findings	Marked erythema and creeping fat (the latter is virtually pathognomonic)	Absent except in severe colitis or toxic megacolon
Perianal complications	Often prominent, including large anal skin tags, deep fissures, perianal fistulas, that are often complex	Not prominent (fissure or fistula if present, should be uncomplicated)
Strictures	Often present	Rarely present; when present, suggests adenocarcinoma
Fistulas	Perianal, enterocutaneous, rectovaginal, enterovesicular, and other	Absent, except for the rare occurrence of rectovaginal or perianal fistula

Feature	Crohn’ s Colitis	Ulcerative Colitis
	fistulas may be present	
Histopathology	Granulomas are present in 15%-60% of patients (higher frequency in surgical specimens than in mucosal pinch biopsies)	Granulomas should not be present (microgranulomas may be associated with ruptured crypt abscess)
	Crypt abscesses may be present	Crypt abscesses and ulcers are the defining lesion
	Focally enhanced inflammation, often on a normal background, is the hallmark	Ulceration on a background of inflamed mucosa
Serology	pANCA in 20%-25%, ASCA in 41%-76%	pANCA in 60%-65%, ASCA in 5%
ASCA, anti-*Saccharomyces cerevisiae* antibody; pANCA, perinuclear antineutrophil cytoplasmic antibody.		

Feldman, M., Friedman, L. S., & Brandt, L. J. (2010).Sleisenger and Fordtran'sGastrointestinal and Liver Disease E-Book: Pathophysiology, Diagnosis, Management. Elsevier Health Sciences

新病人住院後，若懷疑 IBD 應安排什麼檢查？(如何診斷？)

無診斷黃金標準，須盡量排除其他疾病 (IBD mimics)



1. 實驗室檢查應包含： stool routine, stool culture (Salmonella, Shigella, Campylobacter, C. difficile), C. difficile toxin, CMV/EBV IgG/IgM/DNA, Ameba AB, stool Ameba, HIV AB, ANA, anti-ds DNA, HBs Ag, anti-HBs AB, anti-HBc AB, anti-HCV, smooth muscle AB, mitochondrial AB, ANA, anti-Ds DNA, ANCA、CBC、WBC DC、CRP、NA、K、Creatinine, AST, ALT, Bil, ALB. 長期追蹤應注意營養素缺乏問題可適時檢測 Zn, Vitamin B12, Vitamin D, folate, ferritin, Fe, TIBC
2. 進一步內視鏡檢查應做胃鏡及大腸鏡，同時排除上消化道侵犯。若懷疑小腸侵犯則可安排將囊內視鏡或小腸鏡檢查。 病灶需做切片檢查，查看組織學的病理變化，並加染 CMV IHCstain 排除 CMV 感染，但懷疑 toxic megacolon 的病人應避免排大腸鏡。
3. CTE 或 MRE 可以幫忙評估小腸病灶(MANNITOL 使用，應事先告知可能腹瀉之副作用)。

評估嚴重度的常用指標

Crohn’s disease:

Vienna and Montreal classifications for Crohn's Disease.

Clinical factors	Vienna	Montreal
Age at onset	A1: <40 years A2: ≥40 years	A1: below 16 years A2: between 17 and 40 years A3: above 40 years
Disease location	L1: terminal ileum L2: colon L3: ileocolon	L1: ileal L2: colonic L3: ileocolonic L4: isolated upper disease†
Disease behavior	B1: inflammatory B2: stricturing B3: penetrating	B1: nonstricturing, nonpenetrating B2: stricturing B3: penetrating 'p': perianal disease modifier

†L4 is a modifier that can be added to L1–3 when concomitant upper GI disease is present.
'p' is added to B1–3 when concomitant perianal disease is present.
Adapted with permission from [12].

Crohn' s Disease Activity Index (CDAI)

Clinical Readout	Weighting factor
Number of liquid or soft stools each day for seven days	x 2
Abdominal pain(graded from 0-3 on severity) each day for seven days	x 5
General well being, subjectively assessed from 0 (well) to 4 (terrible) each day for seven days	x 7
Presence of complications* *One point each is added for each set of complications: •the presence of joint pains or frank arthritis •inflammation of the iris or uveitis •presence of erythema nodosum, pyoderma gangrenosum, or aphthous ulcers •Anal fissures, fistulae or abscesses. •Fever (> 100 °F) during the previous week.	x 20
Taking Lomitil or opiates for diarrhea	x 30
Abdominal mass (0 as none, 2 as questionable, 5 as definite)	x 10
Absolute deviation of Hematocrit from 47% in men and 42% in women	x 6
% deviation from standard weight	x 1

Remission of Crohn's disease is defined as a CDAI of less than 150.
Severe disease was defined as a value of greater than 450.
Response defined as a fall of the CDAI of greater than 70 points.

Best WR et al, *Gastroenterology* **70** (3): 439–44. 1976.

Harvey-Bradshaw Index of Crohn's disease activity in adults

Patient sense of general wellbeing in the last 24 hours

- ○ Very well (0 points)
- ○ Somewhat below normal (1 point)
- ○ Poor (2 points)
- ○ Very poor (3 points)
- ○ Terrible (4 points)

Patient report of abdominal pain in last 24 hours

- ○ None (0 points)
- ○ Mild (1 point)
- ○ Moderate (2 points)
- ○ Severe (3 points)

Number of liquid stools in last 24 hours

[ ] stools

Finding of an abdominal mass

- ○ No mass (0 points)
- ○ Possible mass (1 point)
- ○ Definite mass (3 points)
- ○ Mass present and tender (4 points)

Complications

- ☐ Arthralgias (1 point)
- ☐ Uveitis (1 point)
- ☐ Erythema nodosum (1 point)
- ☐ Aphthous ulcers (1 point)
- ☐ Pyoderma gangrenosum (1 point)
- ☐ Anal fissure (1 point)
- ☐ Newly discovered fistula (1 point)
- ☐ Abscess (1 point)

0 to 4 points:	Clinical remission
5 to 7 points:	Mildly active disease
8 to 16 points:	Moderately active disease
17 to 100 points:	Severely active disease

Ulcerative colitis:

Frequency of bowel movements
0 = Normal for the patient
1 = 1-2 stools/day in addition to the usual
2 = 3-4 stools/day in addition to the usual
3 = >5 stools/day beyond the usual

Rectal Bleeding
0 = No blood
1 = Blood streaks in less than half of evacuations
2 = Evidence of fresh blood in most of the evacuations
3 = Bowel movements with fresh blood

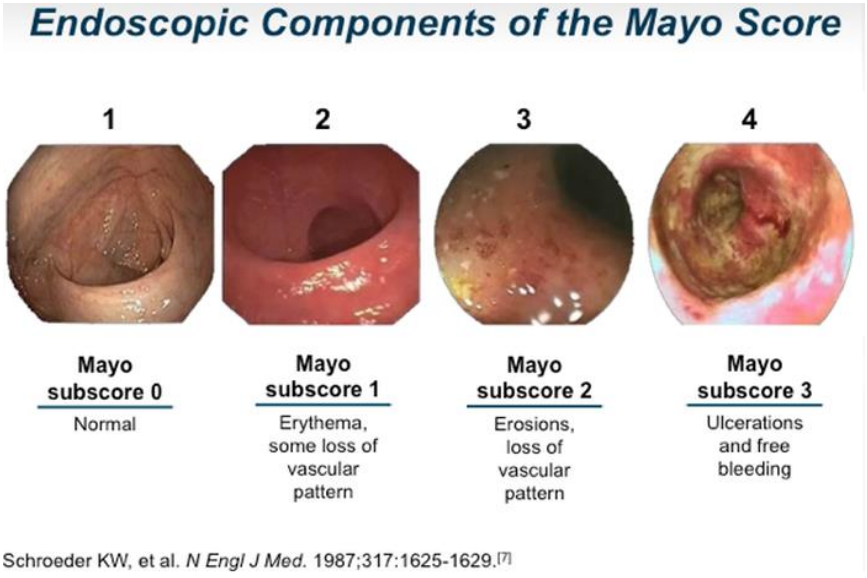
Endoscopic findings
0 = Normal mucosa or inactive disease
1 = Mild disease (enanthema, loss of vascular pattern, mild friability)
2 = Moderate disease (obvious enanthema, loss of vascular pattern, friability, erosions)
3 = Severe disease (spontaneous bleeding, ulceration)

Global Medical Assessment[a]
0 = Normal
1 = Mild disease
2 = Moderate disease
3 = Severe disease

Scores (Points)	Disease severity
≤ 2 and no subscore >1	Clinical remission
3-5	Mild activity
6-10	Moderate activity
11-12	Severe activity

[a]The global medical evaluation takes into account the daily complaint of the patient with regard to abdominal discomfort, pain, a feeling of well-being (normal, above or below the average), physical examination findings and the patient's performance of daily activities.

Mayo score:
Endoscopic Mayo subscore:



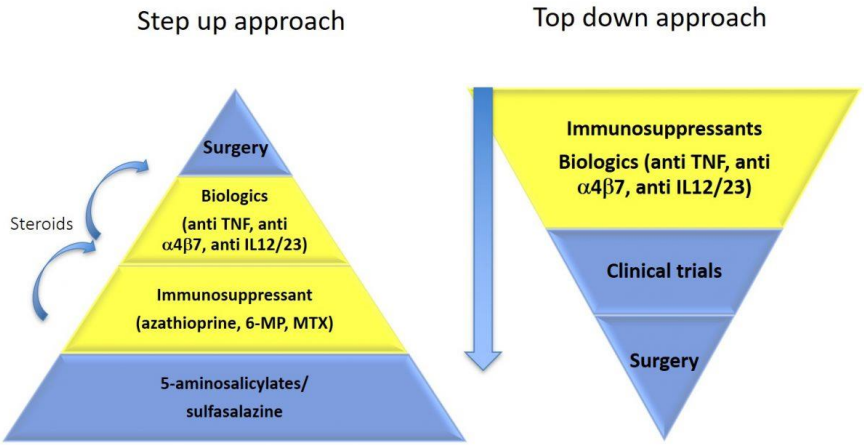
Schroeder KW, et al. *N Engl J Med*. 1987;317:1625-1629.[7]

Truelove and Witts severity index-Ulcerative colitis

	Mild	Moderate	Severe
Bloody stools per day	< 4	4-6	> 6
Pulse	< 90 bpm	≤ 90 bpm	> 90 bpm
Temperature	< 37.5 ℃	≤ 37.8 ℃	> 37.8 ℃
Hemoglobin	> 11.5 gm/dL	≥ 10.5 gm/dL	< 10.5 gm/dL
ESR	< 20 mm/h	≤ 30 mm/h	> 30 mm/h
CRP	Normal	≤ 30 mg/dL	> 30 mg/dL

如何治療：

依照病情嚴重程度治療，目前治療有兩種策略 step up(從藥效較差但副作用少的藥物開始給藥)或是 top down(先給予兩種以上效果較強的藥物再慢慢減量)。若是嚴重病人建議 top down，輕症的病人可以 step up。



治療上藥物選擇:

1. 第一線治療藥物是 5-ASA 這類抗發炎藥物(口服或局部給藥)，可以降低局部發炎，維持長期緩解，但對於 CD 效果不佳。(可能副作用包含頭痛、噁心嘔吐、紅疹)

2. 類固醇則是可以快速緩解發炎，用來做短期緊急治療。由於會造成骨質酥鬆、增加感染風險等，因此盡早(3M)停止類固醇使用，是非常重要的治療目標。

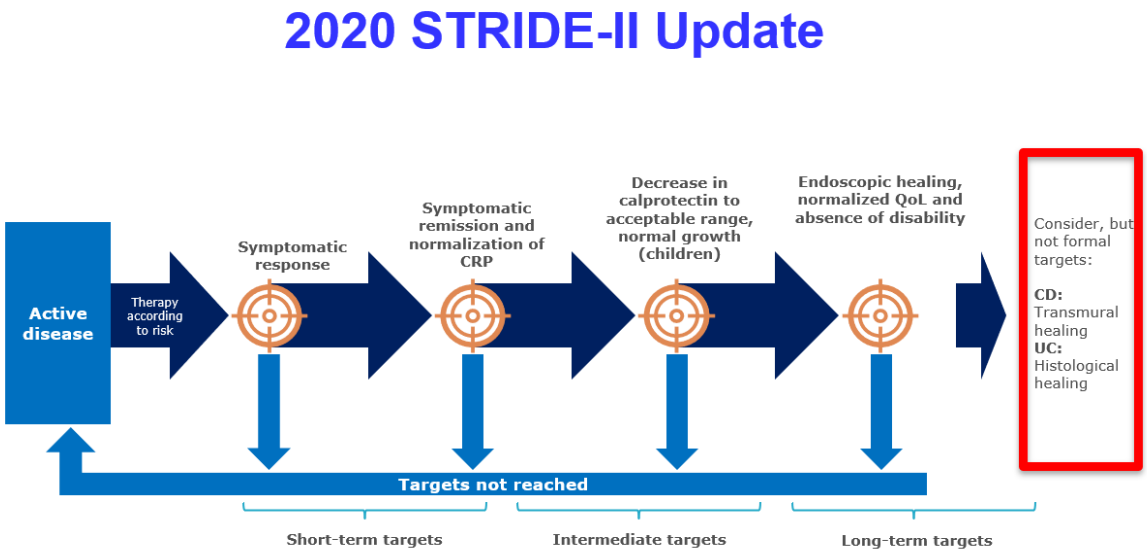
3. 當 5-ASA 無法控制發炎時，則需用進一步使用免疫抑制劑維持長期抑制發炎的作用。 (可能副作用包含頭痛、發燒、噁心、嘔吐、腹瀉、血球降低、感染、肝功能異常)

4. 最後則是生物製劑，一般使用於中重度患者，除了可以控制發炎，減少併發症、避免術後復發，也可以用來治療痲管，在發炎性腸道疾病的治療中扮演舉足輕重的腳色。

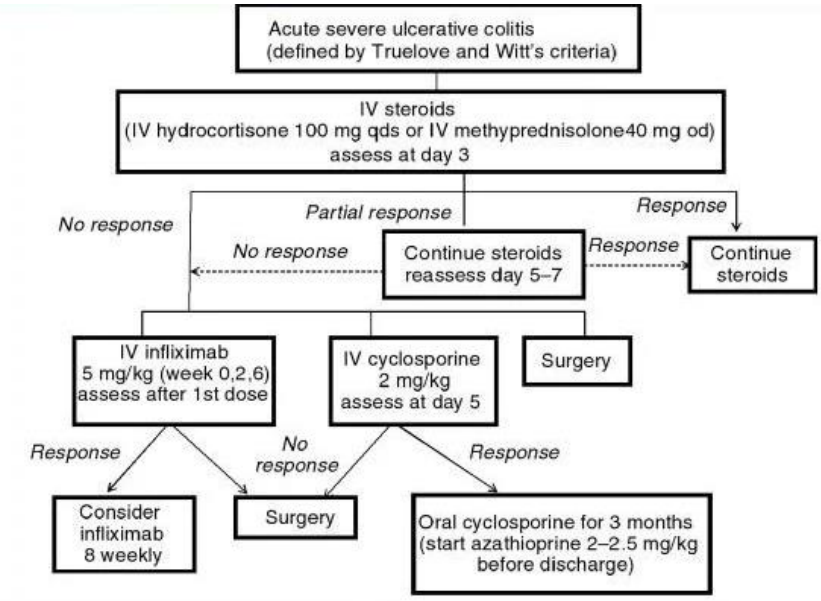
由於生物製劑及小分子藥物價格昂貴，但又是許多中重度患者除了手術之外唯一的選項。因此健保在特定條件下給付生物製劑在發炎性腸道疾病的使用。其中包含多種抗腫瘤壞死因子的生物製劑、IL12/23 inhibitor, JAK inhibitors, 腸道專一性生物製劑。還有許多生物製劑以及小分子藥物仍在不斷發展，許多臨床試驗在進行中，因此在發炎性腸道疾病的治療選擇上日趨多樣性。

治療目標:

目前 IBD 治療目標為黏膜癒合(mucosal healing)，未來 CD 為全層癒合(transmural healing)，UC 為組織學癒合(histological healing)



The treatment of acute severe UC (須保握治療時機)



健保生物製劑使用規範:

目前對於生物製劑於潰瘍性結腸炎之健保給付規範如下，

1.同時符合下列條件；

I. 領有潰瘍性結腸炎重大傷病卡 (直腸型排除)。

II. 經 5-aminosalicylic acid 物、類固醇、及免疫調節劑(如 azathioprine 或 6-mer captopurine) 充分治療無效 (須有病歷完整記載用藥史，連續治療達 6 個月以上)，或對 5-aminosalicylicacid 藥物、免疫調節劑產生嚴重藥物副作用。

III. Mayo score ≧9 分且 Mayo Endoscopic subscore ≧2 分(需檢附兩個月內之大腸鏡報告，內含可供辨識之彩色照片)。

2.急性嚴重的潰瘍性結腸炎，同時符合下列四要件；

I. 內視鏡下符合潰瘍性結腸炎。

II. 病理切片排除巨細胞病毒腸炎、阿米巴結腸炎、淋巴癌。

III. 糞便檢測排除困難梭狀桿菌感染。

IV. Mayo Score 為 12 分，經類固醇全劑量靜脈注射 (如 methylprednisolone 40-60mg/day 等) 連續治療 5 天無效。

3.療效評估與繼續使用：

(1)初次申請生物製劑以 6~8 週為限，治療後達到臨床反應評估者(Mayo Score ≦6 分，且 Mayo Endoscopic subscore ≦2 分)，方得 申請繼續使用。

(2)繼續使用者，需每 16 週評估一次，若評估仍維持前一療程或更低之 Mayo Score 分數，且 Mayo Endoscopic subscore ≦1 分，可 再申請繼續使用 16 週，維持治療以申請兩次為限。

(3)治療 40 週後若病情復發，依初次使用標準再次提出申請。

目前對於生物製劑於克隆氏症治療之健保給付規範如下，

1.診斷為成人克隆氏症，領有克隆氏症重大傷病卡，並符合下列條件之一；

(1)克隆氏症病情發作，經 5-aminosalicylic acid 藥物 (sulfasalazine, mesalamine, balsalazide)、類固醇、及/或免疫抑制劑(azathioprine, 6-mercaptopurine, methotrexate)充分治療超過六個月，仍然無法控制病情(CDAI≧300)或產生嚴重藥物副作用時，且病況不適合手術者。

(2)克隆氏症經 5-aminosalicylic acid 藥物如(sulfasalazine, mesalamine, balsalazide)、類固醇、及免疫抑制劑(azathioprine, 6-mercaptopurine, methotrexate)充分治療超過六個月，或外科手術治療，肛門周圍廔管或腹壁廔管仍無法癒合且 CDAI≧100 者。

(3)克隆氏症經 5-aminosalicylic acid 藥物如(sulfasalazine, mesalamine, balsalazide)、類固醇、及免疫抑制劑(azathioprine, 6-mercaptopurine, methotrexate)充分治療，仍於一年內因克隆氏症之併發症接受二次(含)以上之

手術治療且 CDAI≧100 者。
2.療效評估與繼續使用：
(1)初次申請生物製劑以 8 週為限，治療後，若達到有效緩解之誘導或部份有效緩解之誘導者，方得申請繼續使用。
i 有效緩解之誘導：CDAI≦150 或廔管痊癒 。
ii 部份有效緩解之誘導：CDAI 分數下降≧100 或廔管數量減少。
(2)繼續使用者，需每 16 或 24 週評估一次，評估仍維持前一次療程有效或部份有效緩解時之 CDAI 分數者，方得提出申請續用，每次申請以 16 或 24 週為限。
(3)治療 56 週後須至少再間隔超過六個月後，因病情復發或以其他治療難以控制達上述 3.之(1)(2)(3)之標準才能再次提出申請使用。

須排除使用之情形：
應參照藥物仿單，重要之排除使用狀況包括：
(1)懷孕或正在授乳的婦女。
(2)罹患活動性感染症(active infection)之病患。
(3)未經完整治療之結核病的病患（包括潛伏結核感染治療未達四週者，申請時應檢附潛伏結核感染篩檢紀錄及治療紀錄供審查）。(102/1/1)
(4)惡性腫瘤或具有癌症前兆(pre-malignancy)之病患(但不包括已經接受過充分治療達 10 年以上的惡性腫瘤)。
(5)具高度感染機會之病患：慢性腿部潰瘍、導尿管置留、身上有引流管、人工關節感染，該人工關節尚未摘除者、頑固性或復發性之胸腔感染症病患。
(6)多發性硬化症(multiple sclerosis)。
5.須停止治療的情形：
(1)療效不彰：療效評估未達繼續使用標準者。
(2)其他事項：包括
i 惡性腫瘤
ii 該藥物引起之嚴重毒性 (白血球過低、嚴重過敏)
iii 懷孕(暫時停藥即可)
iv 嚴重間發性感染(暫時停藥即可)。

- **飲食及日常生活注意?**

在第二次世界大戰後的西方工業國家，發炎性腸道疾病發生率不斷上升，因此和食物的關聯性逐漸受到重視，尤其是包含乳製品、高油脂及高糖飲食等所謂的西化飲食。雖然目前在飲食方面，唯一有明確證據的只有喝配方奶的嬰兒，發生發炎性腸道疾病的機率比喝母乳顯著提高，此外並沒有很強的證據說哪些食物

會誘發或加重發炎性腸道疾病。但是，就過去經驗，亞洲國家隨著西化程度愈來愈高，病例數快速上升，因此最好的飲食建議就是回到東方傳統飲食，避免高油高糖及乳製品。因為這畢竟是原本屬於歐美國家的疾病。此外，在腸道黏膜癒合前，需避高纖、刺激性及生魚片等生食，減少發炎及感染風險。

Management of corrosive injury

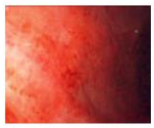
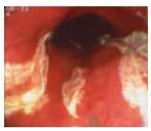
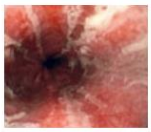
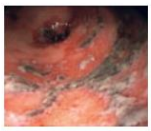
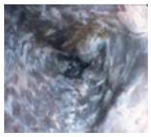
A. Etiology:

1. Acid: Superficial coagulation necrosis
2. Alkali: Liquidfactive necrosis

B. Diagnosis

1. Timing of EGD: preferably within 24 hours (Safely up to 72 hours)
2. Chest x-ray to detection of perforation

C. Zargar’ s grading classification

Grade I	Edema and hyperemia of the mucosa	
Grade IIa	Superficial localized ulcerations, friability, and blisters	
Grade IIb	Circumferential and deep ulcerations	
Grade IIIa	Multiple and deep ulcerations and small scattered areas of necrosis	
Grade IIIb	Extensive necrosis	
Grade IV	Perforation	

D. Management

1. Initially NPO and IV nutrition; On diet until stable condition
2. PPI or H2 blocker use
3. Antibiotics use if infection is suspected
4. Adequate pain control
5. Perforation → Surgery
6. Contraindication: Use of emetics, Neutralizing agents (weakly acidic or basic substances), and NG tube irrigation

E. Complication

1. Stricture: endoscopic dilatation or surgery 3~6 week after injury
2. Cancer: regular follow up
    - ASGE: EGD follow-up 15 to 20 years after the caustic ingestion

Management of foreign body

A. Diagnosis

1. The patient' s history and symptoms and PE for the patient' s general condition and to assess signsof any complications.
2. Radiological evaluation is not recommended for patients with nonbony food bolus impaction without complications.
   - Plainradiography if ingestion of radiopaque objects is suspected or type of object is unknown

Radiodensity	Foreign body
Can mostly be identified on radiography	True foreign bodies (i. e. nonfood objects) Steak bones
C**annot** (regularly) be identified on radiography	Food bolus Fish or chicken bones Wood Plastic Glass Thin metal objects

3. Computed tomography (CT)scan in all patients with suspected perforation orother complication that may require surgery
4. No barium swallow image

B. Type

Type	Examples
Blunt objects	Round objects: coin, button, toy Batteries, magnets
Sharp-pointed objects	Fine objects: needle, toothpick, bone, safety-pin, glass pieces Sharp irregular objects: partial denture, razor blade
Long objects	Soft objects: string, cord Hard objects: toothbrush, cutlery, screwdriver, pen, pencil
Food bolus	With or without bones
Others	Packets of illegal drugs

C. Equipment

Object type	Appropriate retrieval devices
Blunt objects	Grasping forceps, retrieval graspers, polypectomy snare, basket, retrieval net
Sharp-pointed objects	Grasping forceps, polypectomy snare, basket, retrieval net Transparent cap, latex rubber hood
Long objects	Polypectomy snare, basket
Food bolus	Grasping forceps, retrieval graspers, polypectomy snare, basket, retrieval net

D. Timing of FB removal

1. Emergent (preferably within2 hours, but at the latest within 6 hours) therapeutic EGD for foreignbodies inducing complete esophageal obstruction, and for sharp-pointed objects or batteries inthe esophagus.
2. Urgent (within 24hours) therapeutic EGD for other esophageal foreign bodies without complete obstruction or foreign bodies in the stomach such as sharp-pointed objects, magnets, batteries and large/long objects
3. Nonurgent (within 72 hours)therapeutic EGD formedium-sized blunt foreign bodies in the stomach

Object type	Location	Timing
Battery	Esophagus	Emergent
	Stomach/small bowel	Urgent
Magnet	Esophagus	Urgent
	Stomach/small bowel	Urgent
Sharp-pointed foreign body	Esophagus	Emergent
	Stomach/small bowel	Urgent
Blunt and small foreign body < 2 – 2.5 cm diameter	Esophagus	Urgent
	Stomach/small bowel	Nonurgent
Blunt and medium-sized foreign body > 2 – 2.5 cm diameter	Esophagus	Urgent
	Stomach/small bowel	Nonurgent
Large foreign body > 5 – 6 cm	Esophagus	Urgent
	Stomach/small bowel	Urgent
Food bolus	Esophagus	Emergent (urgent if without symptoms or without complete obstruction)

4. Close observation in asymptomatic individuals who have concealed packetsof drugs by swallowing (“body packing”).
    - Endoscopic retrieval isn’ t recommended, and surgical referral is recommended in cases of suspectedpacket rupture, failure of packets to progress, orintestinal obstruction
5. The use of a protective device, such as overtube in order to avoid esophagogastric/pharyngealdamage and aspiration during endoscopic extraction of sharp-pointed foreign bodies.
6. Endotracheal intubation should be considered in the case of high risk of aspiration